

Fusing sequence and graph deep learning considering dynamic influence of built environment for crowd flow prediction

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ABSTRACT

Accurate modeling and prediction of crowd dynamics at a detailed spatiotemporal level is crucial for urban planning and management. Crowd travel is deeply influenced by built environment. But most of the crowd flow prediction research treated the built environment as a static feature. This study proposes a deep learning-based method considering dynamic influence of built environment to overcome the gap. The time-based multiscale geographic weighted regression is utilized to quantify the dynamic impact of built environment on flows over time. Then, a deep learning module is used to capture the spatial and temporal dependency of the crowd flows. Finally, an operation that fuses the spatiotemporal dependency, dynamic impact of built environment and synchronous features is applied to the crowd flow prediction. Two cases in Guangzhou and Beijing are conducted to demonstrate the superiority of the proposed method. The results are the following: 1) in the Guangzhou case, the prediction errors of the proposed method are 18.22 in mean absolute error and 2.97 % in mean absolute percentage error, which outperforms the benchmark methods; 2) in the Beijing case, the method proposed almost has a lower MAE and MAPE than the benchmark methods in 30- or 60-min time interval prediction; and 3) compared with consideration of static impact, the proposed model of the accuracy respectively increases 2.74 % and 5.72 % in mean absolute error and mean absolute percentage error with consideration to the dynamic influence of built environment.

1. Introduction

The population and numbers of various vehicles are increasing rapidly with the development of urbanization. These changes lead to more complex crowd flow conditions that bring greater challenges to public safety (Yang et al., 2023). In the meantime, environmental pollution, such as air (Hilpert et al., 2019) and noise pollution (Wang et al., 2024; Wang et al., 2025b), becomes a more serious problem (Wang et al., 2025a). Therefore, the prediction and characterization of crowd flow is a key foundation for efficient and orderly urban management (Miskolczi et al., 2021).

The performances of the crowd flow prediction methods are influenced by the models' generalized abilities for the input features, such as the spatiotemporal characteristics of past crowd flow and other variables. The spatiotemporal characteristics of past crowd flow play a significant role in predicting future crowd flow (Li et al., 2023a). In the meantime, other variables, such as the built environment (Zhao et al.,

2022a), weather (Zhang et al., 2023a), holidays, peak hours (Guo & Zhang, 2020) and temperature (Zhang et al., 2017) also enhance the prediction accuracy. For example, the built environment-related variables have a great impact on the crowd flow prediction (Tu et al., 2020; Ding et al., 2017). Most studies only considered the static impact of built environments (e.g., POI's density (Jiang et al., 2021), building's floorage (Liu et al., 2022)). Although some research on the dynamic influence of the built environment were implemented, the dynamic influence of the built environment on the crowd flow prediction has not been evaluated. The building properties greatly affect the flow of people. For example, from the perspective of buildings' category, people in residential buildings often commute during the morning and evening. During this period, residential buildings facilitate crowd outflow while simultaneously exerting a certain repelling effect on inflow. And commercial activities in the commercial buildings often attract crowd flow during morning peak hours, and encouraging people to travel during evening peak hours. Besides, more people will flow out or in when the

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building scale is large. Consequently, it is necessary to consider the dynamic impact of built environment (BE) on crowd flow prediction.

A lot of research on the prediction of urban population flow has been conducted in these years. The typical methodologies of crowd prediction were implemented through statistical models, e.g., the Markov model (Adamopoulou et al., 2016) and the Gaussian model (Ding et al., 2021). However, those models do not have high prediction accuracy due to failure to recognize the time-series patterns of crowd flow. With the development of AI technologies, deep learning-based methods have been widely used in the crowd flow prediction thanks to the accurate prediction and efficient data processing process (Li et al., 2019a; Rodrigues et al., 2020). For instance, models such as CNN, LSTM, and GRU are utilized. Some studies incorporate attention mechanisms into temporal models, such as the transformer (Zhang et al., 2022) and LSTM-attention (Cai et al., 2024). There are even considerations of spatiotemporal relationships (Zhao et al., 2022c), such as STGCN and TGCN. But those models almost utilized crowd flow data and meteorological data as input features, neglecting the impact of built environment on crowd flow. Few models (Guo & Zhang, 2020; Jiang et al., 2021) considered such impact, however, these models used static features as input, neglecting spatiotemporal heterogeneity of the impact.

To address these challenges, this study proposes a fusing sequence and graph deep learning considering dynamic influence of built environment (SG-DBE) that captures the complex spatiotemporal dependencies of input features. More importantly, dynamic impact of the built environment on crowd flow, which is quantified by time-based multiscale geographically weighted regression (TBMGWR), is added to a deep learning-based approach based on sequence and graph to further capture invisible dependencies. The contributions of this study are fourfold:

- (1) Recurrent graph attention convolution neural network (RGACNN) is constructed. This deep learning-based approach integrated gated recurrent unit (GRU), graph attention network (GAT) and temporal convolution network (TCN), that adequately captures the spatiotemporal correlations of the input features;
- (2) TBMGWR is introduced into SG-DBE to explicitly capture the spatiotemporal and dynamic correlations between the flow and built environment;
- (3) This study proposes a representation of the randomness of sequences that reasonably reflects the magnitude of randomness between sequences;
- (4) This study calculates crowd flow in traffic analysis zones by integrating the size and direction of both vehicular and pedestrian flows in real-world scenarios.

2. Related work

2.1. Prediction model for crowd flow

The initial research on crowd flow prediction was mostly implemented through time-series analysis methods (Lim & McAleer, 2002; Shahriari et al., 2020; Singh et al., 2020) and their improved methods considering seasonal patterns and multivariate series (Cecaj et al., 2021). These traditional time series prediction methods were conducted based on the stationary assumption of time sequences that led to the deficiencies of failing to model the complex non-linear correlations of space and time (Deng et al., 2017). Moreover, these methods failed to capture the nonlinear and complex correlations of the crowd flows between space and time.

Deep learning-based methods were deployed in later research (Al-Musaylh et al., 2018) to predict the time series-based crowd flow thanks to their abilities of capturing time series-related variables. For example, recurrent neural networks (RNN) (Yuan et al., 2020) were used to extract temporal features individually relying on the memory units. Long short-term memory (LSTM) (Ajitha et al., 2022; Li et al., 2023b)

and gated recurrent unit (GRU) (Li & Zhong, 2023) were widely applied to the crowd flow prediction tasks because of their abilities of capturing the temporal dependence in the crowd flow data. For example, Liu et al. (2018) showed that the LSTM was capable to make the time series-based predictions compared to the statistical (Mo et al., 2022) and traditional machine learning-based models (Hu et al., 2022). He et al. (2019) applied a GRU model to the crowd flow prediction, and the results showed a great performance on the accuracy compared to Autoregressive Integrated Moving Average Model (ARIMA) (Ma et al., 2020) and Seasonal Autoregressive Integrated Moving Average (SARIMA) (Agyemang et al., 2023; Tang et al., 2021). Furthermore, some model incorporating attention mechanisms in the temporal sequence (like Self-attention-LSTM (Cai et al., 2024) and Transformer (Zhang et al., 2022)) was proposed and garnered attention (Cirstea et al., 2022). Although many models considered spatiotemporal correlations, most models used the neural networks based on one principle to capture the temporal correlations. Such neural networks may trigger partial temporal information to be ignored. Thus, some studies have integrated two or more different feature extraction mechanisms to adapt to various temporal dependency patterns, such as CNN-LSTM (Li et al., 2023b), CNN-GRU (Wang et al., 2025a; Ma et al., 2024), CNN-Transformer (Liu et al., 2024; Chen et al., 2024), and LSTM-Transformer (Nascimento et al., 2023). Their performance has proven to be superior to that with a single mechanism to extract features.

Subsequent research has found that the patterns of crowd flow are not only reflected by their own temporal changes but are also to some extent mirrored by the surrounding flow information. Spatiotemporal characteristics, as the non-ignorable elements, have significant impacts on crowd flow predictions (Lin et al., 2019). The extraction of spatiotemporal features of crowd flow can become a necessary condition for models in a multitude of studies. (Guo et al., 2019). Most initial methods considering spatial dependencies utilized convolutional neural networks (CNNs) (Guo & Zhang, 2020). These methods, which treated the data in a grid-based format, achieved great success in exploring complex spatiotemporal dependencies. For example, Liu et al. (2018) proposed a demand prediction model for taxi origin destinations by utilizing two CNNs to learn local spatial dependencies. However, the natural physical connections in the transportation network were disrupted, since the study areas were divided into small grids to meet the input requirements of CNN. To reasonably extract the abundant spatiotemporal features, some studies incorporated the graph structure as a replacement for grids to predict the crowd flow. Graph neural network (GNN) (Liang et al., 2023) was a graph-based neural network inspired by the convolutional layers of CNN. It was adopted to analyze the dependencies of crowd flow on the spatial dimension. Then, GNNs were continuously improved by some researchers. For example, Schlichtkrull et al. (2018) extracted spatially related features through graph convolutional networks (GCN) based on neighboring nodes within the graph structure. Graph Attention Network (GAT) was proposed to focus on a multi-head self-attention mechanism to weigh the relationship between node and its neighbors (Dong et al., 2022). Then, some neural network-based methods considering temporal and spatial dependencies were proposed. Yu et al. (2018) proposed a spatiotemporal graph convolution network (STGCN) integrating GCN and temporal convolutional network (TCN) to respectively capture the spatial and temporal correlations. Zhao et al. (2020) proposed temporal graph convolutional networks (T-GCN), which used GCN to learn the node correlation and extracted the temporal features through GRU. Though those models above consider spatiotemporal correlations, they only apply a CNN-based or RNN-based network to capture the temporal correlations, may causing partial temporal information ignored.

2.2. Build environment and crowd flow

The influence of the urban built environment on the crowd flow has been proved in many research (Chen et al., 2017; Li et al., 2019b; Zhou

et al., 2023). Qian et al. (2015) found taxi demands are largely impacted by trip costs and the urban environment. Population density, commerce, residential area, and a set of time-varying variables, to varying degrees, influence travel demand in Beijing (Li et al., 2019b). Ding et al. (2017) found that built environmental factors like population density and land use will affect the crowd travel. The influence of the built environment on travel demand shows spatiotemporal dynamics (Cai et al., 2020). Spatially, built environments have diverse effects on travel demand in different neighborhoods. Temporally, the same built environment has different effects on travel demand at different periods of the day. Shen et al. (2020) determined the spatiotemporal heterogeneity of travel demand by a geographically and temporally weighted regression (GTWR). The effect of the built environment on travel is subject to spatial and temporal fluctuations. Notably, regression coefficients in temporal distribution exhibited a peak effect.

Various regression-based models have been utilized to identify the impact of building environment on crowd flow. Tin Tin et al. (2012) determined the correlation between crowd flow and the building environment by multiple linear regression (MLR). Li et al. (2020) combined geographic weighted regression (GWR) to explain how the impacts of the building environment factor on the rail transit ridership vary across Guangzhou. Regarding the modeling approaches, existing studies that did not consider scale effects typically applied the models of OLS, generalized linear model (GLM), multiple linear regression, and GWR to uncover the possible controlling determinants of crowd flow (Li et al., 2019a). To address this issue, multiscale geographically weighted regression (MGWR) was developed to allow the scale of each explanatory variable to be altered (Marizal & Monalisa, 2022). In some studies, the built environment-related variables were directly used as input of deep learning model to calculate their influence during the crowd flow prediction. For example, Guo and Zhang (2020) used point of interest (POI) density as an auxiliary feature to enhance the prediction accuracy. Liang et al. (2023) used number point of interest (POI) to assist in predicting travel needs. Yang et al. (2020) utilized temperature, weather, and built environment information as inputs to enhance the accuracy of the prediction of demand for shared bicycle. However, these methods (Ali et al., 2021; Zhang et al., 2023b; Zhao et al., 2022b) fail to identify the spatiotemporal heterogeneity of impact. The impact of BE is considered as static in those models for their static input. In the predictions, the variables related to the built environment inputted into their model essentially remain unchanged, leading to a fixed impact of the built environment on the forecast outcomes. Thus, dynamic impact is necessarily input into a deep learning-based model, which promotes the model to identify comprehensively the impact on pattern of crowd flow. This study integrates the dynamic influences of built environment on crowd flow into a deep graph neural network to improve the prediction performance.

In summary, the performance of a predictive model is related not only to its generalization ability but also to its input features. Research has suggested that crowd flow can be inferred from its spatiotemporal pattern of its historical data but also benefits from building environment. The weaknesses of the related work are the following: 1) not considering the dynamic impact of built environment, and 2) applying deep learning-based model that utilizes a single mechanism to capture the time-based correlations, yet considers spatiotemporal dependencies. Consequently, to address these weaknesses, this study applies time-based multiscale geographic weighted regression (TBMGWR) to obtain the dynamic impact of built environment. On one hand, it can directly quantify the dynamic influence of the built environment on crowd flow; on the other hand, it can be integrated into deep learning models, enabling them to recognize the spatiotemporal variation patterns of the built environment's impact and capture potential nonlinear relationships. Then, RGACNN coupling RNN-based and CNN-based neural network is used to capture the temporal dependencies.

3. Methodology

3.1. Problem description

Definition 1. (Region Division): In this study, traffic analysis zones (TAZs) are used as analysis units that are derived from the study area divided by the truck roads and secondary truck roads.

Definition 2. (TAZ-controlled roads): The road between two intersections is referred to as a road segment, with each segment having two directions. TAZ-controlled roads are considered those road segments directly adjacent to a TAZ in a single direction.

Definition 3. (Graph structure of TAZs): As shown in Fig. 1. Diagram of graph structure, we define the centroid of the building group (with mass as the building area) contained within a TAZ as the center of the TAZ, which corresponds to a node in the graph structure. The two connected nodes must simultaneously satisfy the following conditions: 1) There exists a crosswalk connecting the TAZs corresponding to the two nodes; 2) The crowd flow on any road segment controlled by a TAZ corresponding to one node can enter the road segment controlled by a TAZ corresponding to another node.

Definition 4. (Crowd flow): Crowd flow includes outflow and inflow. They can be described as the following:

$$p_t^i = \sum_j \varepsilon |u_{car,t}^{i \rightarrow j}| + \sum_j |u_{walk,t}^{i \rightarrow j}| \left(j \neq i; i \in Z; j \in Z \cup \tilde{Z}; u_{car,t}^{i \rightarrow j}, u_{walk,t}^{i \rightarrow j} \in u_{k,t}^{i \rightarrow j}; u_{k,t}^{i \rightarrow j} \in U_t \right) \quad (1)$$

$$a_t^i = \sum_j \varepsilon |q_{car,t}^{j \rightarrow i}| + \sum_j |q_{walk,t}^{j \rightarrow i}| \left(j \neq i; i \in Z; j \in Z \cup \tilde{Z}; q_{car,t}^{j \rightarrow i}, q_{walk,t}^{j \rightarrow i} \in q_{k,t}^{j \rightarrow i}; q_{k,t}^{j \rightarrow i} \in Q_t \right) \quad (2)$$

where p_t^i and a_t^i are the crowd outflow and inflow of i -th TAZ at t time interval, U_t represents the collective set of all initiated trips for both cars and pedestrians at t time interval; $u_{k,t}^{i \rightarrow j}$ represent the set that initiate trips by k transport mode (car and walk merely be considered in this study) with the destination j at t time interval; Q_t represents the set of all trips concluded by cars and pedestrians at t time interval. $q_{k,t}^{j \rightarrow i}$ denotes the set that trips concluded by k transport mode with destination i at t time interval; Z represents the set of all Traffic Analysis Zones (TAZs) within the study area, while $\tilde{Z}$ denotes the area outside the study region. Considering that a car may carry multiple passengers, a conversion factor ε needs to be applied when converting traffic flow to crowd flow. “|” indicates the number of elements in set.

Problem 1. (Flow Prediction): In this study, the study area is firstly divided into several analysis units (Zhou et al., 2018; Zhang et al., 2017; Zhang et al., 2018; Ali et al., 2018). Given the feature vectors of past time and feature vector of time interval attribute (e.g. weather, peak hours and weekend (Guo & Zhang, 2020)), the flow prediction problem is to predict the value of crowd flow at the $t + 1$ time interval with multivariate regression function f . It can be described as the following:

$$Y_{t+1} = f(\zeta_{t+1}, \psi_{t+1}) \quad (3)$$

where Y_{t+1} the predictions on the flow (inflow or outflow) of crowd at the $t + 1$ time interval; ζ_{t+1} represent the input of the historical data before $t + 1$ time interval; ψ_{t+1} is feature vector of $t + 1$ time interval.

As Fig. 2 shows, the historical data as input can be divided into two modes(as expressed as Eq. 4): a) short-term data $\zeta_{t+1,k}^S$ (as expressed as

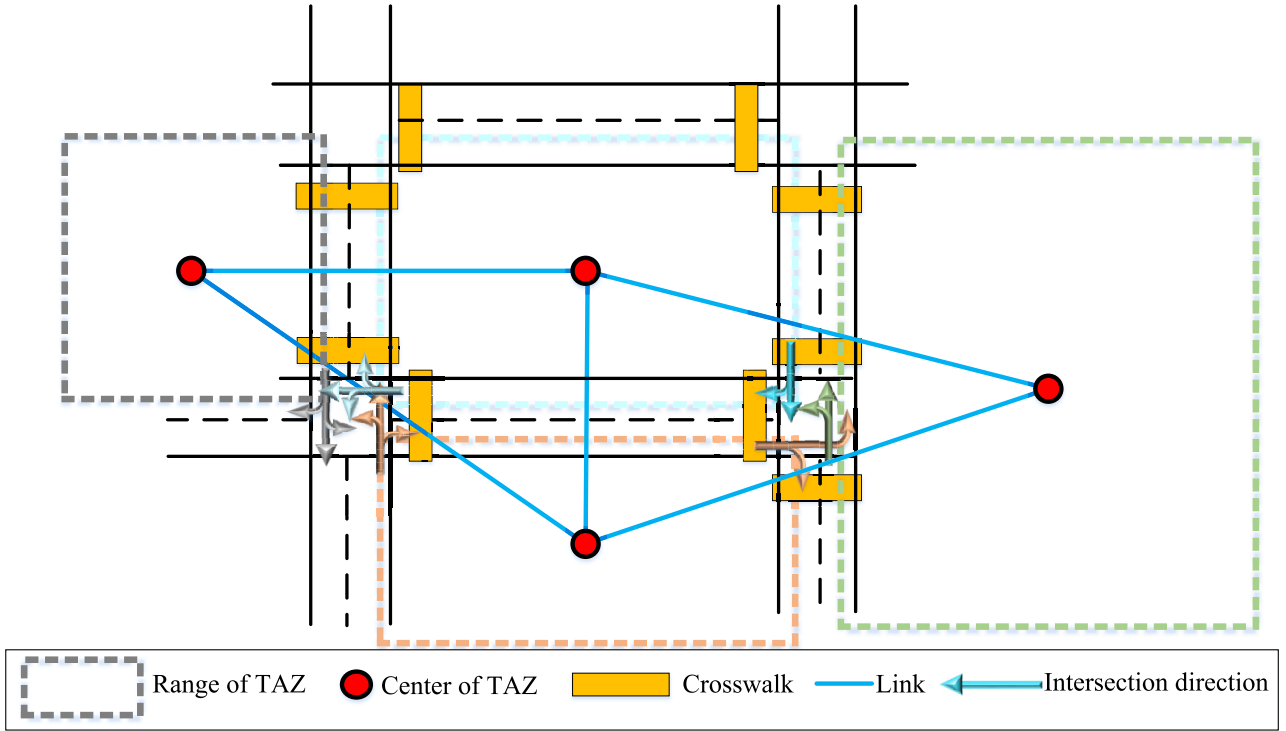


Fig. 1. Diagram of graph structure.

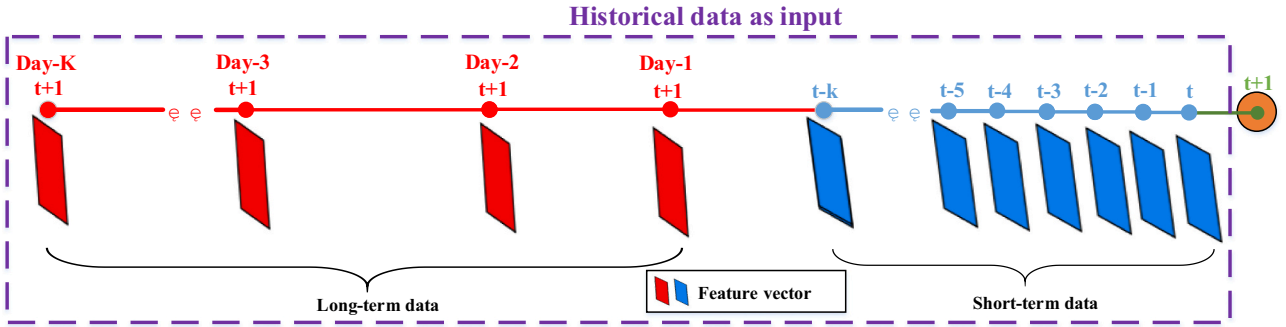


Fig. 2. Diagram of historical data as input.

Eq. 5) includes feature vectors of k time intervals in front of $t + 1$ time interval in d day; b) long-term data $\zeta_{t+1,h}^L$ (as expressed as Eq. 6) includes feature vectors at $t + 1$ time interval of the h days in front of d day. Both of them can be expressed as the Eq. 7.

$$\zeta_{t+1} = [\zeta_{t+1,k}^S, \zeta_{t+1,k}^L] \quad (4)$$

$$\zeta_{t+1,k}^S = [v_t^d, v_{t-1}^d, v_{t-2}^d, \dots, v_{t-k}^d] \quad (5)$$

$$\zeta_{t+1,h}^L = [v_{t+1}^{d-1}, v_{t+1}^{d-2}, \dots, v_{t+1}^{d-h}] \quad (6)$$

$$\zeta_{t+1} = [v_t^d, v_{t-1}^d, v_{t-2}^d, \dots, v_{t-k}^d, v_{t+1}^{d-1}, v_{t+1}^{d-2}, \dots, v_{t+1}^{d-h}] \quad (7)$$

where ζ_{t+1} is the historical data as input for predicting value at the $t + 1$ time interval, including feature vectors of short-term data ($\zeta_{t+1,k}^S$) and long-term data ($\zeta_{t+1,h}^L$); v_t^d denotes the feature vector of d -th day at the t time interval; “[a, b]” denotes concatenation operation to vector a and vector b .

3.2. Overview of the SG-DBE approach

As Fig. 3 shows, there are three steps in the framework of sequence- and graph-based deep learning framework considering dynamic influence of built environment (SG-DBE) proposed by this study. This study collected data about the built environment-related variables including the type of buildings with their floorages at each TAZ, the historical amounts of the inflow and outflow of crowds at different zones and data on time interval attribute. At first, the data of the crowd flow and built environment-related variables are pushed into the TBMGWR to reveal the spatiotemporal influences of built environment on the crowd flow via the product of regression coefficients and floorage. The crowd flow, historical influences of building environment are input into the RGACNN, which integrates temporal convolution network (TCN), gated recurrent unit (GRU), that are used to capture the temporal dependence, and graph attention network (GAT) that is utilized to capture the spatial dependence. Data of time interval attribute is pushed into embedding layer for reducing the dimensionality. Finally, all the hidden features are fused in fusion layer, and output to predict the crowd flows in each TAZ.

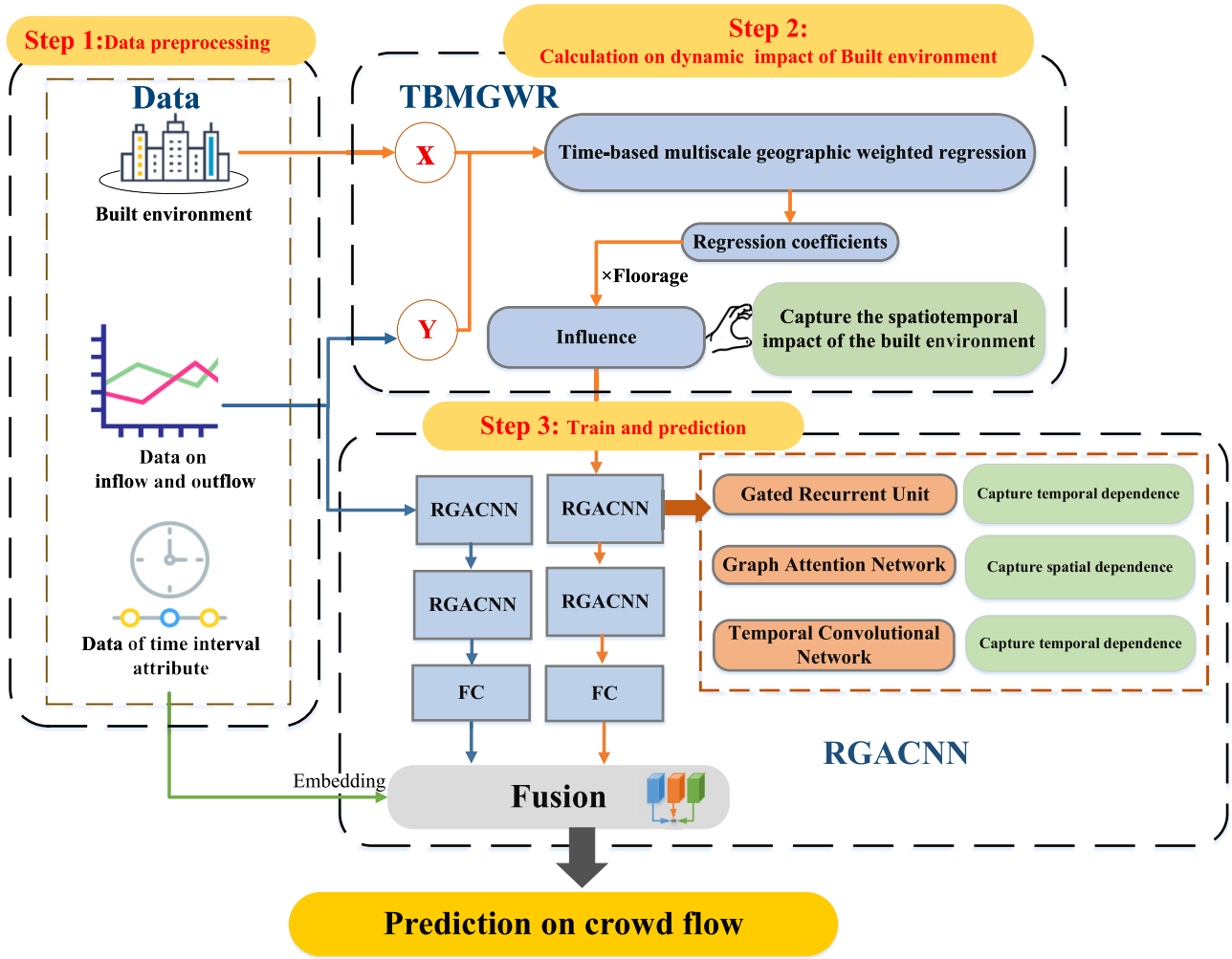


Fig. 3. The framework of SG-DBE.

The details are described in the following sections.

3.3. Time-based multiscale geographic weighted regression

In this study, time-based multiscale geographic weighted regression (TBMGWR) is proposed to capture the spatial relationships in time series, overcoming the disadvantage of temporal heterogeneity neglected by multiscale geographic weighted regression (MGWR). The function is given by following:

$$p(a)_{t+1}^i = \beta_{t+1}^o(u_i, v_i) + \sum_m \beta_{t+1, b_m}^m(u_i, v_i) x_{mt}^i + \varepsilon_{t+1}^i \quad (8)$$

where $p(a)_{t+1}^i$ is the dependent variable referring to the crowd outflow (inflow) of the i -th TAZ at the $t + 1$ time interval; (u_i, v_i) is the coordinates of the center of the i -th TAZ (centroid belonging to the set of buildings); x_{mt}^i is the independent variable referring to the floorage of m -th type of building of the i -th TAZ at the t time interval; $\beta_{t+1}^o(u_i, v_i)$ is the regression constant of the i -th TAZ at the t time interval; $\beta_{t+1, b_m}^m(u_i, v_i)$ is the regression coefficient based on the optimal bandwidth b_m of the corresponding independent variable m ; ε_{t+1}^i is the random error of the model.

The one-dimensional parameter matrix $\tilde{\beta}_{t+1}(u_i, v_i)$ including regression constant and regression coefficients can be expressed by the Eq. 9.

Then, it can be approximately calculated by weighted least squares method shown in Eq. 10.

$$\tilde{\beta}_{t+1}(u_i, v_i) = \left[\beta_{t+1}^o(u_i, v_i) \quad \beta_{t+1, b_1}^1(u_i, v_i) \quad \beta_{t+1, b_2}^2(u_i, v_i) \quad \dots \quad \beta_{t+1, b_m}^m(u_i, v_i) \right]^T \quad (9)$$

$$\tilde{\beta}_{t+1}(u_i, v_i) = (X_t^T W(u_i, v_i) X_t)^{-1} X_t^T W(u_i, v_i) P(A)_{t+1} \quad (10)$$

where $P(A)_{t+1}$ is a one-dimensional vector (as expressed by Eq. 11) for crowd outflow (inflow) at $t + 1$ time interval; X is an independent variable matrix (as expressed by Eq. 12); $W_t(u_i, v_i)$ is spatial weight matrix (as expressed by Eq. 13); n is the number of TAZ.

$$P(A)_{t+1} = \left[p(a)_{t+1}^1 \quad p(a)_{t+1}^2 \quad p(a)_{t+1}^3 \quad \dots \quad p(a)_{t+1}^n \right]^T \quad (11)$$

$$X_t = \begin{bmatrix} 1 & x_{1t}^1 & x_{2t}^1 & \dots & x_{mt}^1 \\ 1 & x_{1t}^2 & x_{2t}^2 & \dots & x_{mt}^2 \\ 1 & x_{1t}^3 & x_{2t}^3 & \dots & x_{mt}^3 \\ \dots & \dots & \dots & \dots & \dots \\ 1 & x_{1t}^n & x_{2t}^n & \dots & x_{mt}^n \end{bmatrix} \quad (12)$$

$$W_t(u_i, v_i) = \text{Diag}(w_t^{i,1}, w_t^{i,2}, w_t^{i,3}, \dots, w_t^{i,n}) = \begin{bmatrix} w_t^{i,1} & 0 & 0 & \dots & 0 \\ 0 & w_t^{i,2} & 0 & \dots & 0 \\ 0 & 0 & w_t^{i,3} & \dots & 0 \\ 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & w_t^{i,n} \end{bmatrix} \quad (13)$$

The bi-square function considers only the sample points that are within a bandwidth b from the regression point, assigning a weight of 0 to all other sample points. This approach enhances computational efficiency. Thus, the distance weights w_t^{ij} in the $W_t(u_i, v_i)$ can be represented as Eq. 14 (Tang et al., 2023).

$$w_t^{ij} = \begin{cases} \left(1 - \left(\frac{d_{ij}}{b_t}\right)^2\right)^2 & (d_{ij} \leq b_t) \\ 0 & (d_{ij} > b_t) \end{cases} \quad (14)$$

where w_t^{ij} is the distances weight from i -th TAZ to j -th TAZ based on the

bandwidth b to independent variable x at the t time interval; d_{ij} is distances between i -th TAZ to j -th TAZ; b_t is the bandwidth to independent variable x at t time interval. This study uses a back-fitting algorithm (Goetz et al., 2011) to maximize expected logarithmic likelihood for multi-scale bandwidth b search.

3.4. Recurrent graph attention convolution neural network

As shown in Fig. 4, Three types of input features are used in RGACNN. They are respectively crowd flow data (A), historical influence of building environment (B) and data of time interval attribute (C), which can be considered as tensor. Those tensors have three dimensions (as shown in Fig. 5): length of sequence, number of nodes and number of features. According to the definition of the historical data as input, the features of A can be represented by Eq. 15, with a feature length of $k + h$. The features of B can be represented by Eq. 16, with a feature length of $(k + h) \times m$. The features of C can be represented by Eq. 18 with a feature length of j . Those three tensors are respectively conducted normalization operations and divided into multiple batches firstly to accelerate training speed and enhance training quality. Then, A and B after

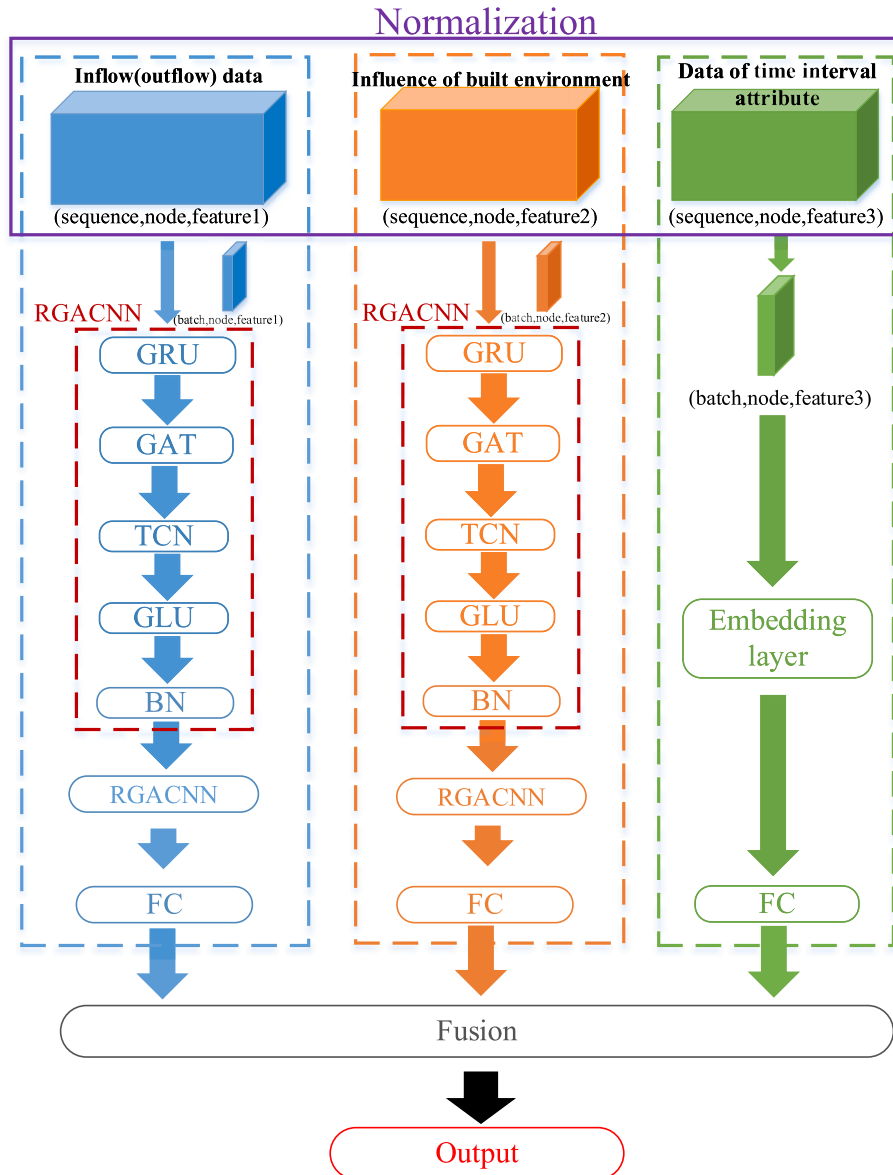


Fig. 4. The framework of RGACNN.

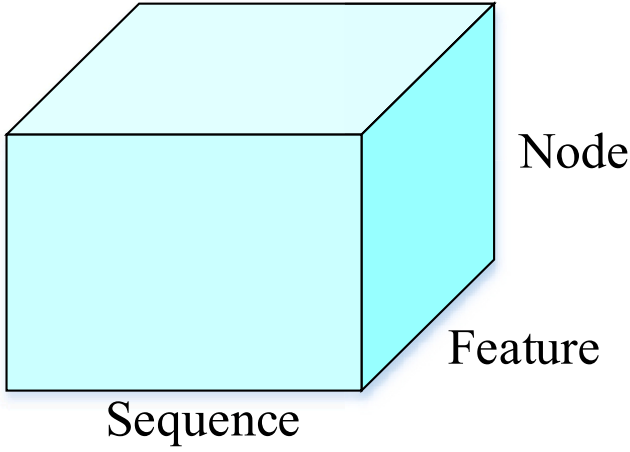


Fig. 5. The diagram of three-dimensional tensor

processing are input into the Recurrent graph attention convolution neural network (RGACNN) for processing. RGACNN module that integrates with GRU, GAT, TCN, GLU, BN. In where, GRU (RNN-based) are used to capture the temporal correlation of input features. Then, GAT is used to capture spatial dependencies of input data. To minimize the loss of features during the spatial feature extraction process, and considering the temporal dependencies of crowd flow, a Temporal Convolutional Network (TCN) is subsequently employed after the Graph Convolutional Network (GCN) to further extract temporal features. And it can make the model adapt to complex temporal dependencies. Besides, C is input into an embedding layer to reduce the dimensionality of the features while capturing the similarities between them. Finally, all of three are pushed into full connection layer (FC) to conveniently conduct feature fusion. Details of each module are described in the following sections.

$$\zeta_{t+1}^A = [p(a)_t^d, p(a)_{t-1}^d, p(a)_{t-2}^d, \dots, p(a)_{t-k}^d, p(a)_{t+1}^{d-1}, p(a)_{t+1}^{d-2}, \dots, p(a)_{t+1}^{d-h}] \quad (15)$$

$$\zeta_{t+1}^B = [\rho_t^d, \rho_{t-1}^d, \rho_{t-2}^d, \dots, \rho_{t-k}^d, \rho_{t+1}^{d-1}, \rho_{t+1}^{d-2}, \dots, \rho_{t+1}^{d-h}] \quad (16)$$

$$\rho_t^d = [\rho_{t,1}^d \times S_1, \rho_{t,2}^d \times S_2, \rho_{t,3}^d \times S_3, \dots, \rho_{t,m}^d \times S_m] \quad (17)$$

$$\psi_{t+1}^C = [\ell_{t+1}^1, \ell_{t+1}^2, \ell_{t+1}^3, \dots, \ell_{t+1}^j] \quad (18)$$

where $p(a)_t^d$ is the value of outflow(inflow) at the t time interval on d day; ρ_t^d is the influence matrix of building environment at the t time interval on d day; $\rho_{t,m}^d$ is the regression coefficient calculated by TBMGWR; S_m is the floorage of the m -th type building; ℓ_{t+1}^1 is the m -th attribute of $t+1$ time interval.

3.4.1. Spatial features extraction

To extract spatial dependences behind input features, graph attention network (GAT) is applied in RGACNN. GAT (Liu et al., 2018) is a graph-based node classification neural network coupling attention mechanism that considers different degrees of importance between the node and its neighboring node. In GAT, each node in the graph can be assigned corresponding weights based on the features of its neighboring nodes. A self-attention mechanism is applied to the nodes to evaluate the importance between center node i and its adjacent node j by the attention coefficient α_{ij} (Eq. 11).

$$\alpha_{ij} = \frac{\exp(\text{LeakyReLU}(a^T([Wh_i, Wh_j])))}{\sum_{k \in \text{Node}(i)} \exp(\text{LeakyReLU}(a^T([Wh_k, Wh_k])))} \quad (19)$$

where h_i and h_j respectively represent the features of i and j , referring to the features mapped to i and j after TCN; W is the weight matrix for input linear transformation; a^T is a weight vector that parameterizes the attention mechanism, that is a single-layer feedforward neural network focusing on the most relevant parts of the input data; “[a, b]” denotes the concatenation operation between vector a and vector b .

To enhance the model’s capability of capturing spatial features, a multi-head self-attention mechanism is incorporated. It can be expressed by the following:

$$h'_{i,R} = \sigma \left(\frac{1}{R} \sum_k a_{ik}^R W^R h_k \right), \quad k \in \text{Node}(i) \quad (20)$$

where a_{ik}^R is the attention coefficient derived from the R -th attention mechanism; W^R is the corresponding weight matrix; $\text{Node}(i)$ is the set of i ’s adjacent nodes; σ is the nonlinear activation function.

3.4.2. Temporal features extraction

To make GAT capture spatial feature by considering a measure of temporal dimension, this study uses gated recurrent unit (GRU) (Cho et al., 2014) to preliminarily capture temporal dependences of data. GRU is a more effective structure for LSTM (Irie et al., 2016). The GRU layer (Wang et al., 2023) has two main components: the update gate (r_t) and the reset gate (z_t). The two gates are obtained after an activation function σ applied to the weighted sum of X_t and h_{t-1} . The reset gate regulates the degree to which status information from the previous time step ignored. Then, the update gate determines the portion of historical information that should be incorporated into the current status. By the two gates, a current memory status (k_t) is used to store the associated information from historical data and compute the current hidden status (h_t). The relevant expressions can be given as Eq. 13–16.

$$Z_t = \sigma(W_z x_t + U_z h_{t-1} + b_r) \quad (21)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (22)$$

$$\tilde{h}_t = \tanh(W_h x_t + r_t \otimes U_h h_{t-1} + b_h) \quad (23)$$

$$h_t = (1 - z_t) \otimes h_{t-1} + z_t \otimes \tilde{h}_t \quad (24)$$

where x_t and h_t are the input and hidden state at the t time interval; W_z , W_r , W_h , U_z , U_r , and U_h are the affine transformation parameter matrices; “ $\otimes$ ” represents the Hadamard product (Cho et al., 2014); b_z , b_r , and b_h represents the corresponding biases; $\tanh$ is the nonlinear activation function.

Temporal convolution network (TCN) follows the GAT to handle the temporal relationships of data. TCN is an improvement of CNN. The causal convolution is represented by Eq. 20. For each layer of a convolutional network at time t , the weight update p only originates from the weight changes of its previous layer at time t or earlier.

$$y_t = \prod_{i=1}^t P(x_t | x_1, x_2, \dots, x_{t-1}) \quad (25)$$

Dilated Convolution (given by Eq. 26) is a data processing method to capture the relationship between historical inputs and current outputs and improves processing efficiency by extending convolution.

$$F(s) = \sum_{i=0}^{k-1} f(i) \cdot x_{s-dt} \quad (26)$$

where F represents the dilated convolution; s is the element of the sequence; x is the 1-D sequence input; f is a filter with k size; d is the dilation factor; $s - dt$ accounts for the direction of the past.

Residual connection (He et al., 2016, 2019) solves the gradient vanishing problem through skipping connections. The residual module is shown as Eq. 27.

$$o_t = x_t + \text{Relu}(W_2(\text{Relu}(W_1 x_t)) + b) \quad (27)$$

where x_t, o_t are the input and output data; w_1, w_2 are the weighted matrix; b is the deviation; Relu is the activation function.

3.4.3. Activation function of RGACNN

Relu activation function- and Sigmoid activation function-based Gated Linear Unit (GLU) is generated through the gate unit to address the issue of gradient explosion and vanishing during gradient back-propagation (Yann et al., 2016). It can be described as the following:

$$h = (W_1 * X + B_1) \otimes \sigma(W_2 * X + B_2) \quad (28)$$

where w_1, w_2 are the parameter matrixes; X, h are the input and output data; B_1, B_2 are the biases.

3.4.4. Fusion of features and full connection

The fully connected layer (FC), as a linear transformation, is used to extract useful features from the input data h through the parameter matrix W and deviations b between two layers. The output data Y of FC is given, as shown in Eq. 29.

$$Y = Wh + b \quad (29)$$

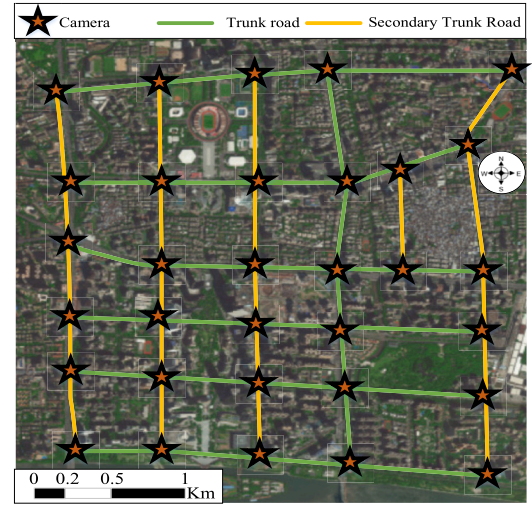
The operation of feature fusion aims to fuse the three outputs in the same size from FC layer each time with parametric-matrix-based fusion. Specifically, the output of the fusion is given by the Eq. 30.

$$Y_{t+1} = \sigma(W_{\text{flow}} \otimes h_{\text{flow}} + W_{\text{reg}} \otimes h_{\text{reg}} + W_{\text{ts}} \otimes h_{\text{ts}}) \quad (30)$$

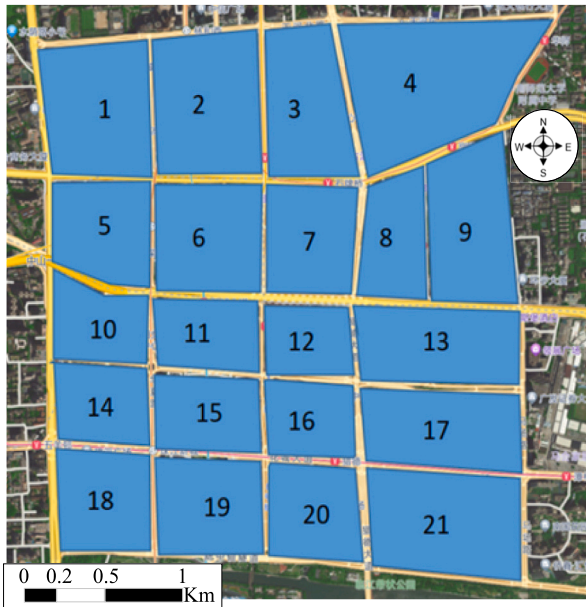
where $h_{\text{flow}}, h_{\text{reg}}, h_{\text{ts}}$ are the output of FC layer belonging to three type of features, $W_{\text{flow}}, W_{\text{reg}}, W_{\text{ts}}$ are the corresponding parameter matrix.



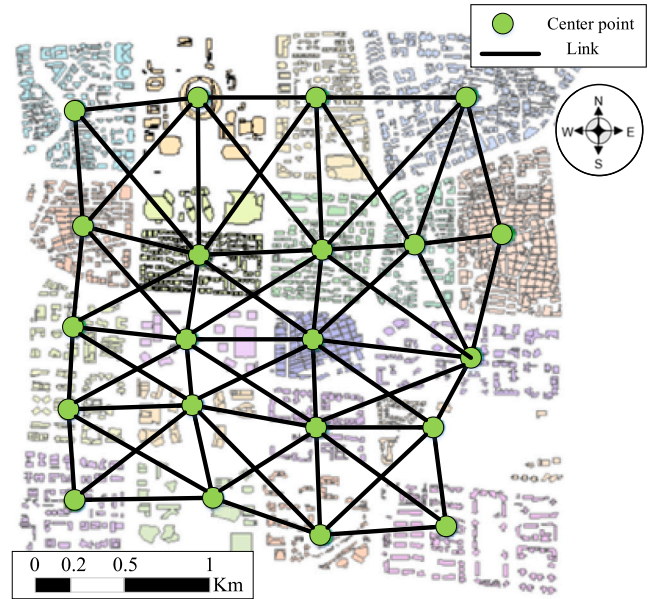
(a) Size of study area



(b) Coil, camera and road division situation



(c) Illustration of all TA



(d) The graph structure of study area

Fig. 6. Information of study area.

4. Case study

4.1. Data description and preprocessing

As illustrated in Fig. 6(a), the research area selected for this study was a certain part of Guangzhou, and the area was divided into 21 TAZs based on the main and secondary truck roads (shown in Fig. 6(c)). The camera monitoring of intersections along the main and secondary roads (shown in Fig. 6(b)) recorded the vehicular flow data and pedestrian flow data were obtained continuously for 76 days. Then, this study extracted the data of buildings' attributes including floorage and coordinate and the center of each TAZ from Gaode Map (Zhao et al., 2022a), and drew the graph structure of this area according to definition 3 (shown in Fig. 6(d)).

The selection of a 15-min interval is because the minimum stable duration of traffic flow is at least 15 min. In one day, there can be 96 time slots with intervals of 15 min each, ranging from [00:00–00:15, 00:15–00:30, 00:30–00:45, ..., 23:45–24:00]. The initial data, as shown in Table 1, requires information including Date ID, Time, Intersection ID, Camera ID, and Vehicle ID.

The planar information of intersections is depicted in Fig. 7. Each intersection determines the direction of vehicle travel based on the sequential Camera IDs of the same Vehicle ID at that intersection. In Appendix B, the data collection and preprocessing based on two modules are presented definitely.

The dataset after preprocessing had three parts including the data of crowd flow in each TAZ, the built environment-related data, and the data of time interval attribute. The details of this dataset are shown in Table 2.

The built environment-related data included the type, floorage, and coordinates of the buildings. The whole buildings were classified into 12 types based on the related regulation (Wang et al., 2025a), including the Residence, Administration, Education, Sports, Medical care, Commerce, Comprehensive office, Recreation, Park, Cultural, Supply and Squares. The other data used in this study included the time sequence. Partial missing values in the data were filled through the interpolation method (Zhao et al., 2022b). This study multiplied the filled values by a random fluctuation coefficient (i.e., 5 % in this study) to demonstrate a certain degree of randomness. The dataset was distributed in 60 % for training, 10 % for validation and 30 % for testing based on time series. The crowd flow and built environment-related data were normalized with the range of [0,1] through the min-max normalization method (Guo & Zhang, 2020). Besides, the one-hot encoding method (Zhang et al., 2017) was used to process the data of time interval attribute, such as 0 being used to represent the time interval is within peak hour or weekend, and 1 being used to represent the time interval is within off-peak hour or weekday.

4.2. Experimental setup

Through facial recognition from camera imagery, we have tallied that on average, each vehicle carried 1.2 individuals during the data collection period. According to the definition in Eq. 1, the conversion

factor ε is set to 1.2.

For the convenience of model building and prediction accuracy, with considering the limitation of computing power, the batch length is set at 32. All weight parameter matrixes in the network are optimized by minimizing the mean square error of the Adam optimizer for 1000 epochs, And the initial learning rate is set as 0.015 with a decay rate of 0.95 after every 30 epochs. The time steps of short-term data and long-term data for the input features are respectively 9 and 7. What's more, the RGACNN module's number is set at 2; the number of GRU layer, the size of convolution kernel and dilation factor in TCN and Size of multi-head in GAT are set at 1, 3, 2 and 3, respectively. All experiments are conducted on the supercomputer Tianhe-2 and a machine equipped with an NVIDIA GeForce RTX 3060 Laptop GPU and 11-th Gen Inter(R) Core (TM) i7-11800H@2.30 GHz CPU.

4.3. Preliminary analysis

4.3.1. Fluctuation of crowd flow

The range of temporal fluctuations in crowd flow, the trend and periodicity, are key aspects that the time series prediction models aim to capture. The degree of randomness in temporal fluctuations increases the difficulty of prediction. Fig. 8(a) and (b) respectively represent the statistical features of crowd outflow for the 1st and 15th TAZ that separately represent two common crowd flow characteristics including weak stationarity and non-stationarity. The 1st TAZ shows a smaller fluctuation range. Its change trend is similar to the change trend of the mean curve. However, it is different from the 15th TAZ's change trend with the regular and small-scale variance changes.

The fluctuation range, mean, and variance of the first-order difference of TAZs' crowd flow is used to measure the degree of randomness in the time series, especially the randomness of the fluctuation trend. As Fig. 9 presents, the fluctuation range of the first-order difference with weak stationarity is much smaller than that with non-stationary. It reveals that the fluctuation trend of the non-stationary time series has relatively greater randomness for the range scale of variance with much smaller than that with non-stationary, which will increase the difficulty of prediction.

4.3.2. Dynamic spatiotemporal influence of the built environment

The regression coefficient derived from TBMGWR is used to reflect the size and direction of the effect on flow by floorage per unit for different types of buildings. Therefore, the regression coefficient is multiplied by the building floorage to measure the impact of different types of buildings within TAZ on flow. Take two typical TAZs (the 1st and 19th) as examples. Fig. 10(a)–(b) respectively show the temporal variations of the built environment's impact on outflow and inflow of 1st TAZ. Fig. 10(c)–(d) respectively show the temporal variations of the built environment's impact on outflow and inflow of 19th TAZ. It can be observed that for both outflow and inflow volumes, the impact of various types of buildings approaches zero around 4:30 p.m. Residential, commercial, and comprehensive buildings have a more pronounced impact compared to other types, which is attributed to the fact that the study area is a mixed commercial and residential zone where these types of buildings cover a larger area and significantly influence crowd travel. The impact trend of change also reflects the travel habits of the population to a certain extent. For instance, Fig. 10(a) and (c) illustrate that the influence of residential buildings on outflow begins to exert a positive effect around 4:30 p.m., indicating the commencement of people's outgoing activities. And the impact intensifies with the morning peak hour approaching; the impact diminishes during the evening peak hour. Conversely, Fig. 10(b) and (d) shows that the impact of residential buildings on inflow increases during the evening peak hour, reflecting the travel activities of people returning home from work.

Fig. 11 shows the impact of the built environment on flow also varies from the spatial locations. The influences of residence at the 10th, 11th, 12th and 15th TAZs are weak, while the influences of commerce are

Table 1
Initial data.

Data ID	Date	Time	Intersection ID	Camera ID	Vehicle ID	...
22412	2016/03/20	12:00:05	5012	K218	S2311xxxx	...
...	...	...	...	...	...	...
22421	2016/03/20	12:00:06	5012	K217	S2311xxxx	...
...	...	...	...	...	...	...
22721	2016/03/20	12:41:06	5025	K241	S2311xxxx	...
...	...	...	...	...	...	...

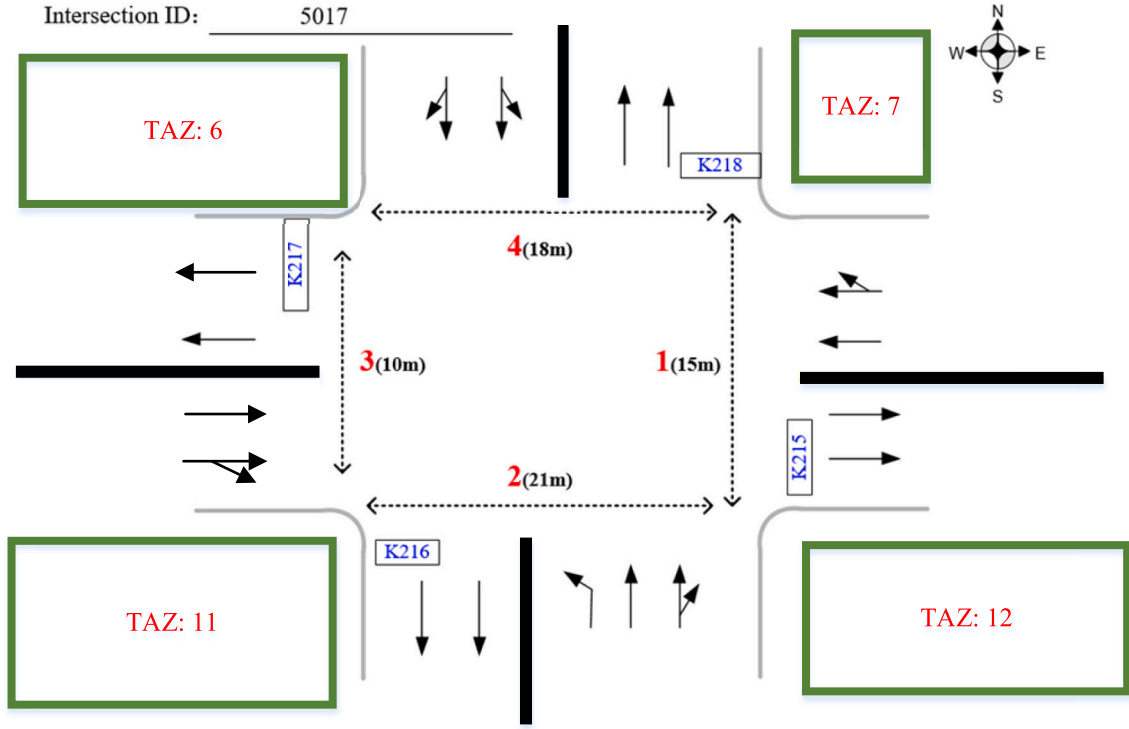


Fig. 7. The planar information of intersections.

Table 2

The descriptions for the dataset after processing and cleaning.

Title of data and its item	Description
Traffic flow data at intersection	
Timespan	Feb. 12, 2016–Apr. 30, 2016 (total 76 day)
Traffic flow	[Flow, TAZ ID, Direction, Time] (e.g. [253, 12, 1, 16:00–16:15], 0 for outflow, 1 for inflow)
Time interval	15 min
Pedestrian flow data at intersection	
Timespan	Feb. 12, 2016–Apr. 30, 2016
Pedestrian flow	[Flow, TAZ ID, Direction, Time] (e.g. [141, 12, 1, 16:00–16:15], 0 for outflow, 1 for inflow)
Time interval	15 min
Built environment data	
Types of buildings	Residence, Administration, Education, Sports, Medical Care, Commerce, Comprehensive office, Recreation, Park, Cultural, Supply, Squares.
Attribute of buildings	Floorage and coordinates of each building
Data of time interval attribute	
Time serial number	Serial number (e.g. [1, 2, 3, ..., 96, 1, 2, 3, ...])
Type of time interval	0 for the off-peak time interval, 1 for the peak time interval with the time of 6:30 a.m.–8:30 a.m. and 4:30 p.m.–8:30 p.m.
Weekend	1 for the Weekend, 0 for the weekday

relatively stronger at these TAZs that are commercial and business districts. The 4th, 5th, 6th, 7th, 9th, 18th, 19th, 20th and 21st TAZs are mixed commercial, business and residential areas, which explains the positive and negative influences on inflow and outflow in morning peak hour. Those impacts represent a certain clustering phenomenon. Therefore, it is necessary to consider these dynamic spatiotemporal influences of the built environment for the crowd flow prediction.

4.3.3. Spatiotemporal dependencies of crowd flow

The correlation between outflow (inflow) at t th time interval and the same time interval for 5 days is calculated through Pearson's correlation (Chen et al., 2023). And the results show that the correlation coefficients between them are above 0.96 with a P -value of 0. What's more, as shown in Fig. 12, a trend of decreasing correlation with the advance of time periods is presented. It can be found that the correlation coefficients between t and $t - 1$ to $t - 6$ are above 0.8. Moreover, for outflow, the correlation between t and its last time interval reaches 0.99. These correlations verify that the crowd flow holds a strong temporal dependency.

Global Moran's index is used to calculate the spatial dependence of crowd flows through the outflow and inflow of each TAZ at the time intervals in peak hours. The results (shown as Fig. 13) suggest that crowd flows have a significant spatial auto-correlation in peak hours, which is consistent with the related research (Zhao et al., 2022). and present an upward trend at morning peak hours.

4.4. Result of crowd flow prediction

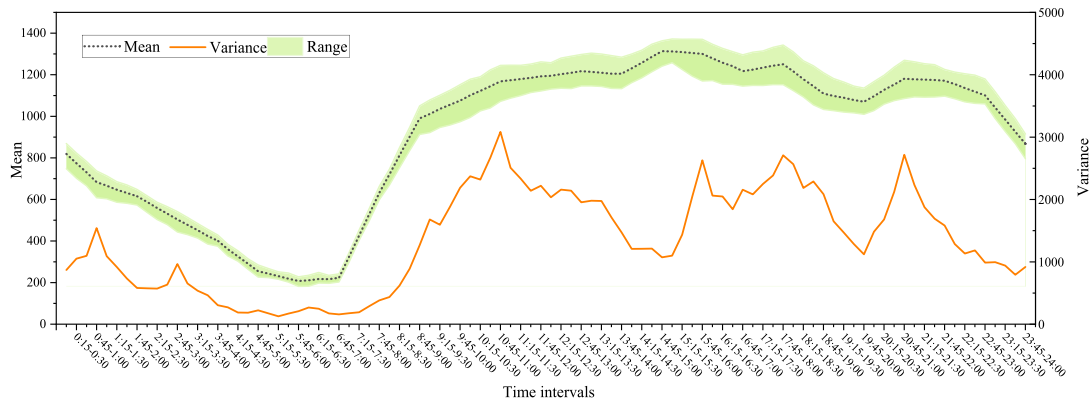
The performance metrics included Mean Absolute Percentage Error (MAPE) (Moreira-Matias et al., 2013), Mean Absolute Error (MAE) (Ke et al., 2017), and Acceptance Rate (ATR) in this study. These metrics are defined as below:

$$\text{MAPE} = \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^N \left| \frac{y_{i,t} - \hat{y}_{i,t}}{y_{i,t}} \right| \quad (31)$$

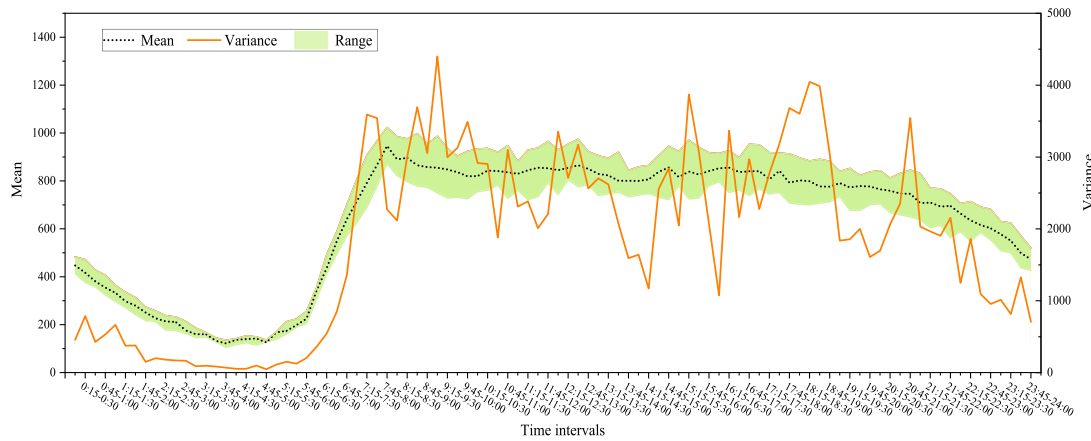
$$\text{MAE} = \frac{1}{T} \sum_{t=1}^T \sum_{i=1}^N |y_{i,t} - \hat{y}_{i,t}| \quad (32)$$

$$\text{ATR}(R) = \frac{\text{count}_{\text{ARE}, R\%}}{T \times N} \quad (33)$$

where $y_{i,t}$, $\hat{y}_{i,t}$ are the real and prediction of the i th TAZ at t time interval, respectively; $\text{count}_{\text{ARE}, R\%}$ is the number of the absolute relative error



(a) 1-stTAZ



(b) 15-th TAZ

Fig. 8. The distribution of range, mean and variance of crowd outflow over time.

(ARE) within $R\%$; T , N are the length of sequences and the amount of TAZs, respectively.

4.4.1. Analysis of Model Performance

Eight benchmark methods, as shown in Table 3, are used to compare with the proposed model (For detailed information regarding the parameters of the deep learning models used in this study, please refer to Appendix A).

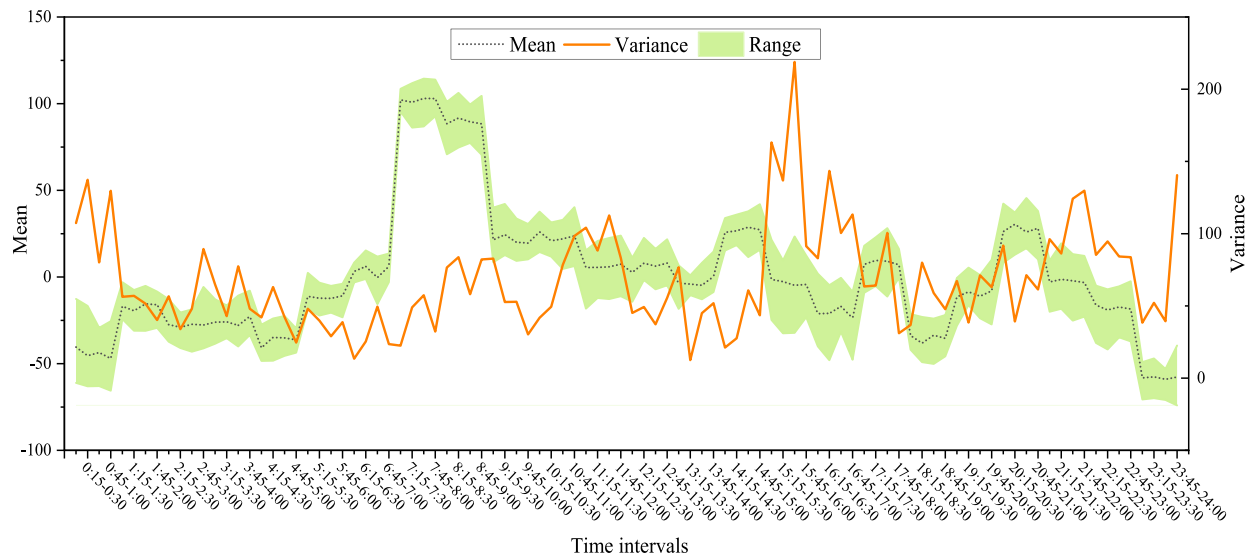
The training time of SG-DBE on conventional a computer is 3.8 h, while the training time on the Supercomputer Tianhe-2 is 4.2 min. Table 4 shows the average performance of the proposed model and benchmark models on the testing data in the 21 zones. It can be found that the proposed SG-DBE model achieves the best performance for crowd flow prediction in terms of MAE, MAPE and ATR. The conventional models, such as ARIMA, achieve unsatisfactory prediction accuracy in MAPE and ATR, and even have worse results than quantitative statistical methods (HA). This indicates that the presence of nonlinear relationships in crowd flow significantly affects the forecasting ability of ARIMA models. The RNN-based models, such as LSTM and GRU get better results with the MAE of 21.22, 21.95 and the MAPE of 3.58 %, 3.65 %, while they have better prediction performances than HA and ARIMA with considering the temporal dependency. However, their results are slightly inferior to the GCN-based models (e.g., T-GCN, STGCN, LSGCN, SG-DBE) due to their inability of capturing feature information in space. The TCCT leverages the superior attention mechanism in the transformer and the powerful CNN's dimensionality and feature extraction capabilities, compensating for its deficiency in spatial feature

extraction, ultimately achieving excellent predictive performance. However, The SG-DBE combines two temporal feature extraction mechanisms, demonstrating excellent spatiotemporal extraction capabilities, and also emphasizes the dynamic impact of buildings. The combination of these two strengths results in performance superior to other benchmark methods.

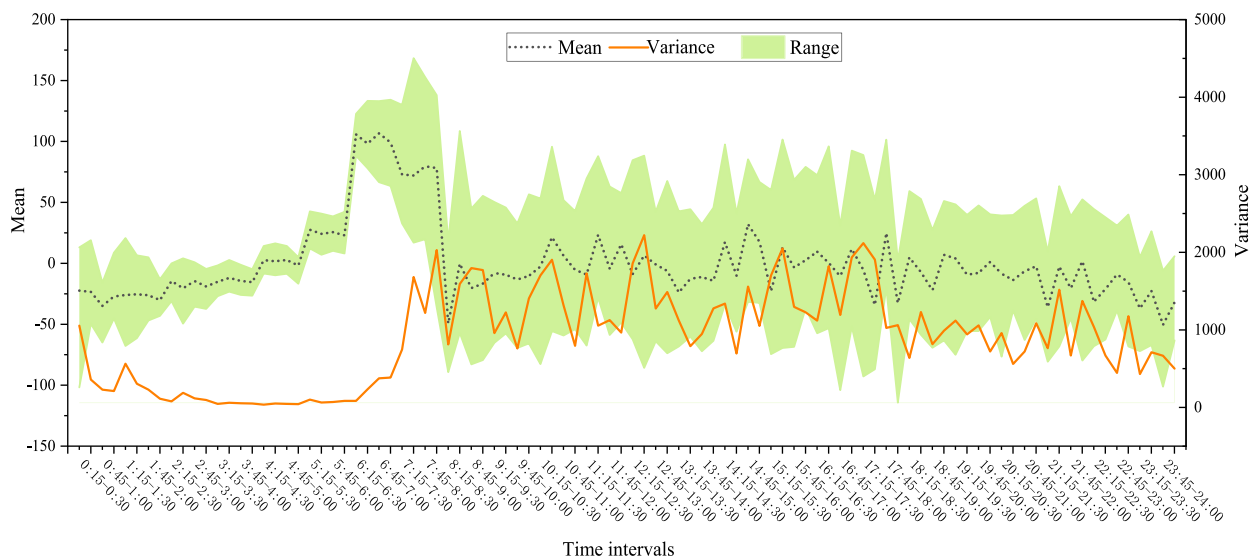
To further analyze the differences on error statistics between the proposed model and benchmark methods, this study separately obtains their error distributions in whole day, morning peak hours and evening peak hours. As Fig. 14(a)-(f) show, the distributions of relative error and residual for SG-DBE are more concentrative based on the length of the boxplot. A smaller range of 1.5IQR of SG-DBE can indicate a lower presence of outliers. Consistent with what is expressed in Table 4, the error distributions of ARIMA and HA are large in the three time periods and also have a greater number of outliers. The error distributions of LSTM and GRU are also larger than those of STGCN, TGCN, TCCT, LSGCN, and SG-BE. The position of the median value can roughly indicate the central location of the error distribution, the median of the errors for SG-BE is nearly concentrated at 0 in the three time periods, implying that most of the errors are converging towards 0, indicating good model performance.

4.4.2. Importance of the dynamic impact of building environment

To verify the importance of the dynamic impact of building environment, we choose different combinations of features (crowd flow data (A); dynamic influence of built environment (B); data of time interval attribute (C)); static influence of built environment (D) like floorage of



(a) 1st TAZ



(b) 15th TAZ

Fig. 9. The distribution of range, mean and variance of crowd outflow's first-order difference over time

building) as the input of the SG-DBE. Details are shown in Table 5.

As shown in Table 6, using the built environment-related and data of time interval attribute in the crowd flow prediction is feasible based on the decreased MAE (MAPE) of 8.73 % (9.43 %), 2.58 % (1.68 %) and 6.59 % (8.08 %) compared to the SG-DBE (A), SG-DBE (A + B) and SG-DBE (A + C).

This demonstrates the importance of input features to model performance. What's more, SG-DBE has a declined MAE(MAPE) of 2.74 % (5.72 %) compared to SG-DBE (A + D + C), and a declined MAE(MAPE) of 4.39 % (7.40 %) compared to SG-DBE (A + D). It demonstrates the advantages of consideration on the dynamic influence of building.

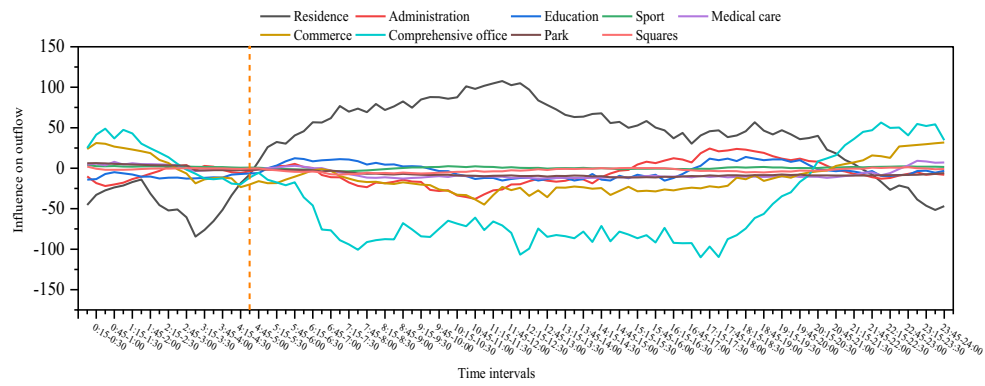
Furthermore, to further validate the positive impact of the dynamic built environment on other model frameworks, additional dynamic built environment influences were incorporated as inputs into three benchmark models (LSTM, T-GCN, and STGCN), with the specific structure detailed in Appendix C. The prediction errors are presented in Table 7 with the values in parentheses indicating the degree of improvement compared to the original models. It can be observed that the dynamic

built environment influences have varying degrees of positive effects on the prediction results of the three benchmark models. Specifically, the positive impact on LSTM is quite weak, while T-GCN and STGCN show considerable improvements. This is attributed to the fact that T-GCN and STGCN are better at extracting the spatiotemporal features of the built environment influences compared to LSTM. It is evident that the extent of the positive effect of the dynamic built environment influences on the prediction structure is related to the feature extraction capabilities of the neural network models.

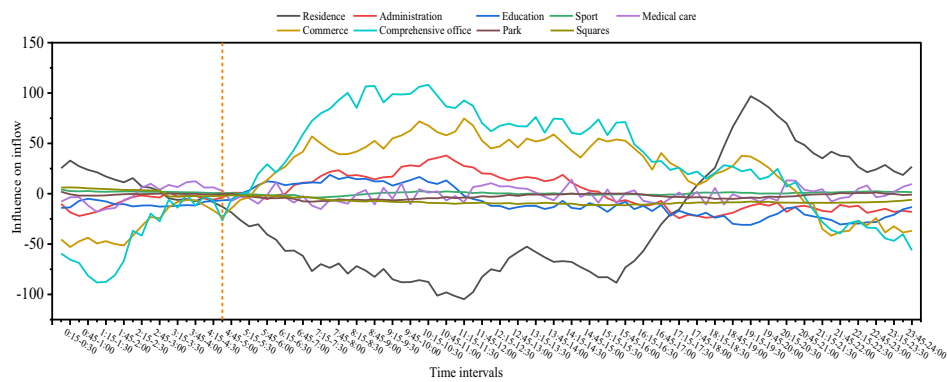
4.4.3. Error analysis

In Fig. 15, the temporal distribution of Relative Errors (RE) for crowd flow prediction in 21 TAZs are presented. Most absolute relative errors are below 5 %, and the error distribution is relatively uniform. Afterward, this study selected the 1st, 3rd, 15th, and 9th TAZs with the smallest to largest MAPE (1.39 %; 2.38 %; 3.54 %; 5.57 %) to analyze their prediction errors (shown as Fig. 16).

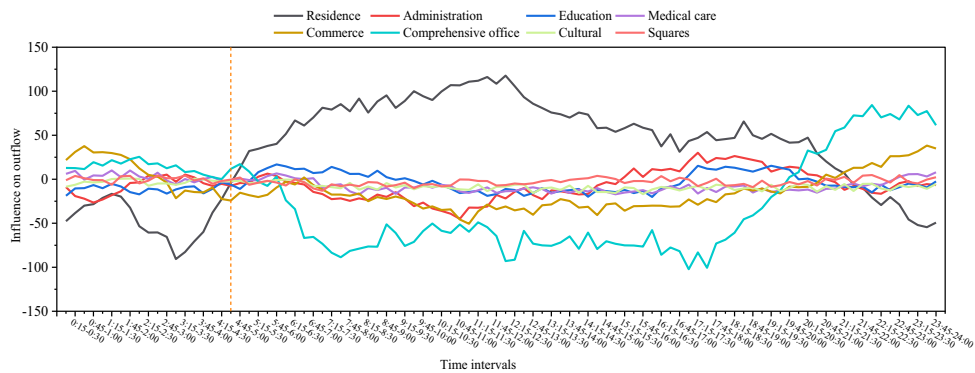
The relative errors increase at the peak-to-valley of crowd flow (as



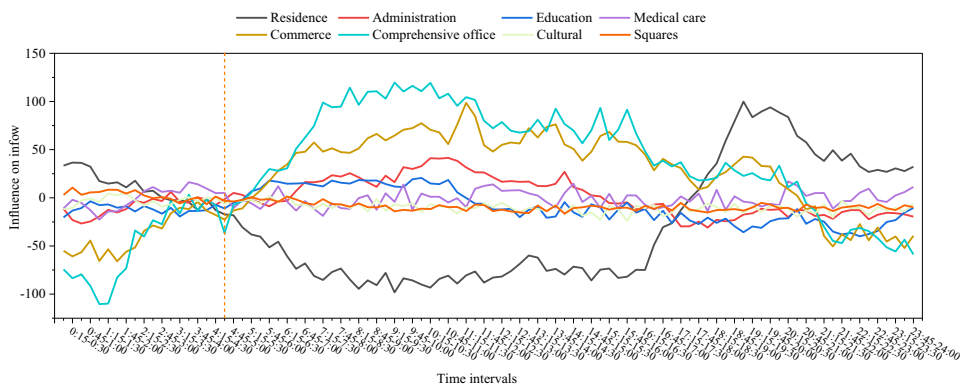
(a) Influence on outflow in 1-st TAZ



(b) Influence on inflow in 1-st TAZ



(c) Influence on inflow in 19-th TAZ



(d) Influence on inflow in 19-th TAZ

Fig. 10. Temporal variation in the influences on crowd flow of the 1st and 19th TAZs.

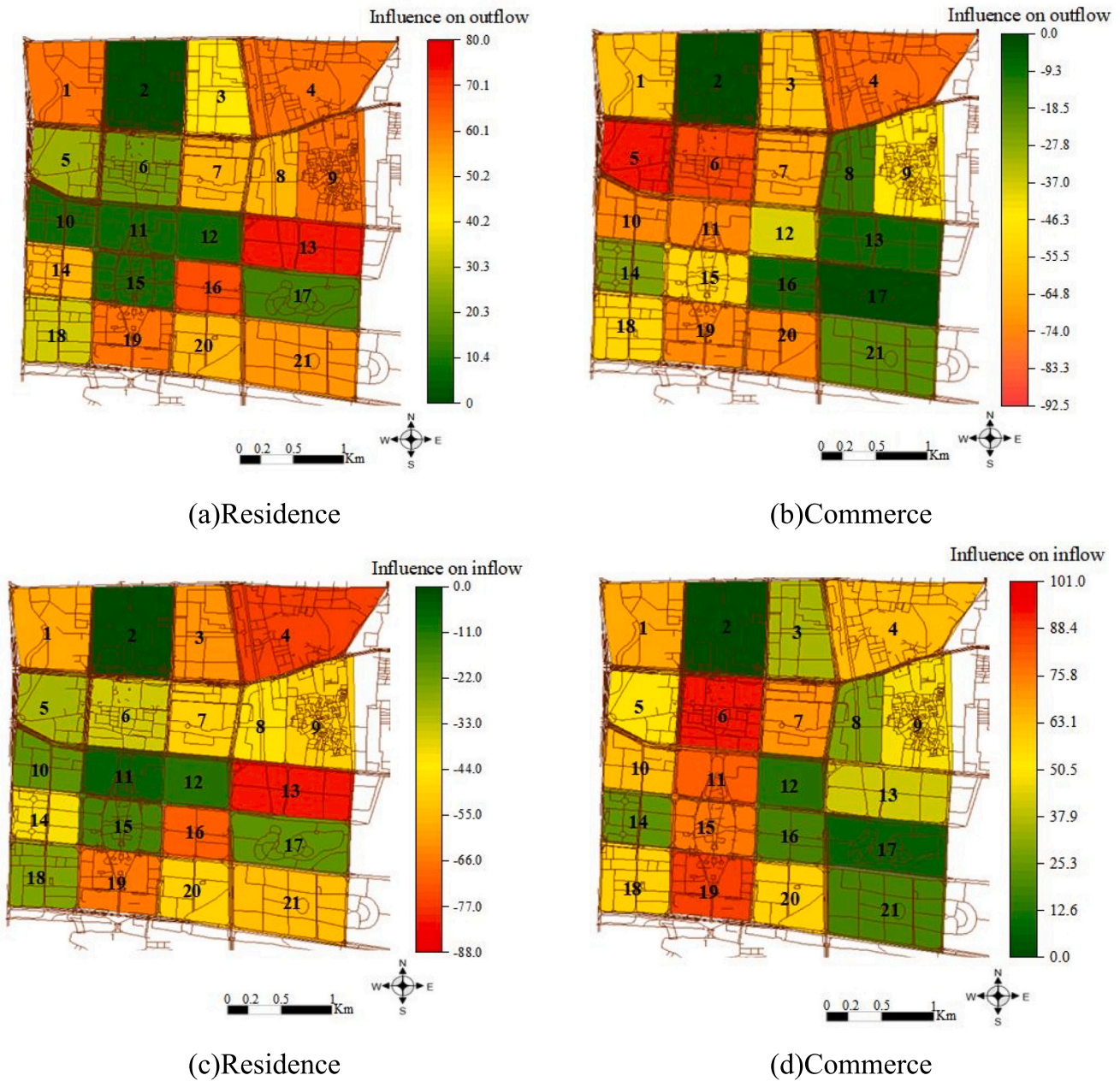


Fig. 11. Spatial distribution of the average influence at the morning peak (6:30 a.m.–8:30 a.m.).

Fig. 16 shows), because the scale of the residuals at the low-flow time periods is closer to the scale of actual flow when compared to other time periods. Its residual is relatively smaller compared to other time periods. Besides, a better predictive performance has the smoother the curve and the smaller the fluctuations. The randomness of the curve's fluctuations is quantified to analyze this phenomenon.

To better quantify the randomness in temporal fluctuations, we considered the variance of flow in each time period of a day as well as of first-order difference for the flow, with the purpose of respectively measure the range of temporal fluctuations and the magnitude of the fluctuation trend. The randomness can be described as the following:

$$C^i = \frac{(V_1^i \times V_2^i) - (V_1^i \times V_2^i)_{\min}}{(V_1^i \times V_2^i)_{\max} - (V_1^i \times V_2^i)_{\min}} \quad (34)$$

$$V^i = \sqrt{\frac{\sum_{t=1}^T \kappa_t^i V_t^i}{T}} \quad (35)$$

$$\kappa_t^i = \frac{V_t^i}{\sum_{k=1}^T V_k^i} \quad (36)$$

where C^i represents the randomness of the time sequence for flow in i th TAZ; V_1^i and V_2^i denote the weighted standard deviation of flow and its first-order difference in i th TAZ; V_t^i and κ_t^i denote the variance of flow in i th TAZ at t time interval and its weight; T is length of time sequences.

To avoid evaluating the predictive performance of each TAZ erroneously due to flow scale issues, we used the natural breakpoints method to classify the TAZs based on the average flow of themselves, aiming to have similar scales within each category. The results are as follows: 1)

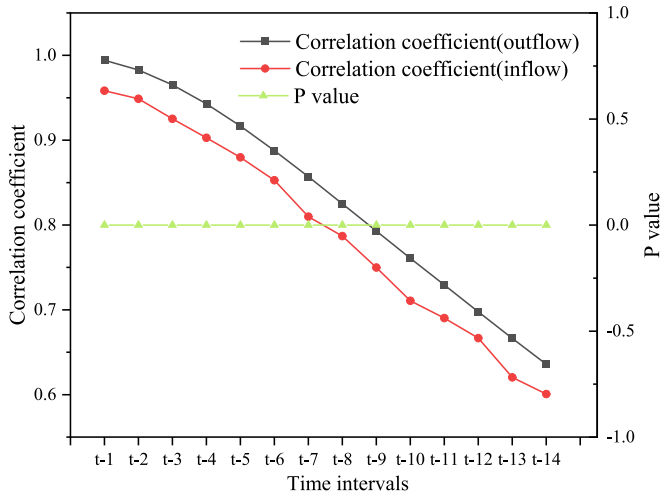


Fig. 12. Temporal dependency of crowd flow.

Category A: 10, 12, 14, 16, 17; 2) Category B: 1, 2, 3, 4, 5, 8, 9, 11, 13, 15, 18, 19, 20, 21; The relation between randomness of the time sequence for flow and prediction performance is shown in Fig. 17. It can be seen that MAPE shows an overall upward trend with C increases, proving that large fluctuations in randomness can affect prediction accuracy to some extent.

4.5. Application of other datasets

To further validate our model's effectiveness, we chose a Beijing area (Fig. 18(a)) for verification. Based on the definitions, we divided it into 39 TAZs (Fig. 18(b)) and determined the adjacency matrix according to the graph structure (Fig. 18(c)). We used ArcGIS to classify and count the buildings in the study area, and the building data preprocessing followed the methods in Section 4.1.

The experiments were conducted on the TaxiBJ open-source dataset (Zhang et al., 2017), as described in Table 8. Considering that vehicle flow can serve as a proxy for crowd flows, we leveraged vehicle trajectory statistics to quantify vehicle flows across TAZs. Specifically, we accumulated crowd flow data in 30-min and 60-min intervals. To ensure a fair comparison, the parameter settings of our model were identical to those in Section 4.2, and the benchmark models along with parameters

were consistent with those in Section 4.3. The results are shown in Table 9.

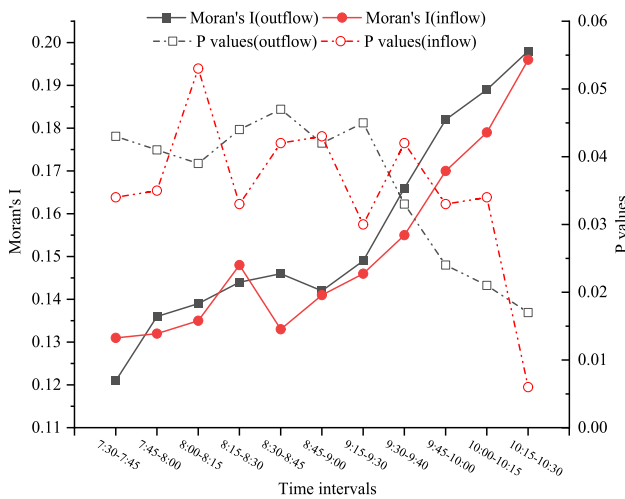
Compared with benchmark methods, our SG-DBE outperforms almost them on the datasets. For example, in 15 min time interval, our SG-DBE reduces by 4.92 % MAE and by 2.62 % on TaxiBJ dataset than the LSGCN model which performs the best among the eight models

Table 3
Benchmark methods.

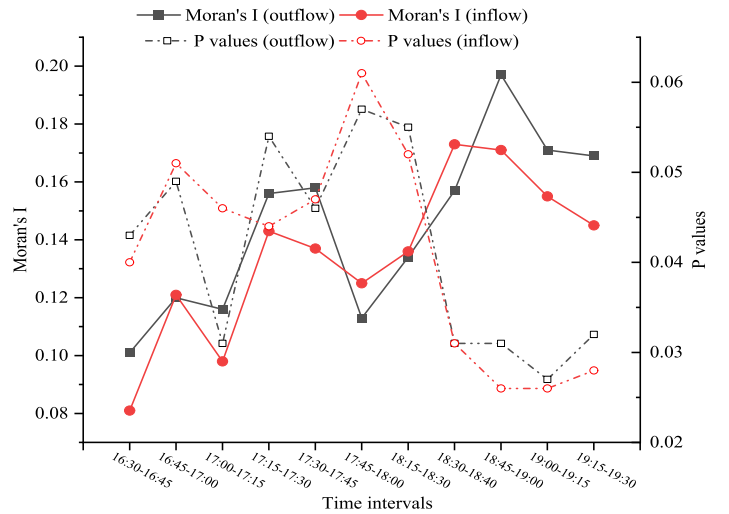
Method	Description
1 HA	Historical average
2 ARIMA (Box & Pierce, 1970)	Autoregressive Integrated Moving Average model: a time-series model for prediction.
3 LSTM (Irie et al., 2016)	Long short-term Memory Network: a variant of RNN.
4 GRU (Cho et al., 2014)	Gate Recurrent Unit: A variant with faster computing efficiency of LSTM.
5 T-GCN (Zhao et al., 2020)	Temporal graph convolutional network: GCN for capturing spatial dependencies, and GRU for capturing temporal dependencies.
6 STGCN (Yu et al., 2018)	Spatiotemporal Graph Convolutional Network: it integrates graph convolution and gated temporal convolution through the spatiotemporal convolutional block for traffic prediction.
7 TCCT (Shen & Wang, 2022)	The model combines Transformer and CNN tightly.
8 LSGCN (Huang et al., 2020)	Long Short-term Graph Convolutional Networks. It presents a new graph attention network called CosAtt.

Table 4
Comparison for prediction performance between proposed model and benchmark models.

Model	MAE	MAPE	ATR (5;10)
HA	54.40	5.62 %	68.87 %; 88.04 %
ARIMA	48.71	8.58 %	42.77 %; 72.32 %
T-GCN	20.45	3.43 %	78.17 %; 95.89 %
LSTM	21.22	3.58 %	76.97 %; 95.17 %
GRU	21.95	3.65 %	75.52 %; 95.73 %
STGCN	20.98	3.38 %	78.12 %; 95.13 %
LSGCN	19.31	3.25 %	81.11 %; 96.47 %
TCCT	18.52	3.06 %	81.75 %; 97.06 %
SG-DBE	18.22	2.97 %	82.03 %; 97.19 %

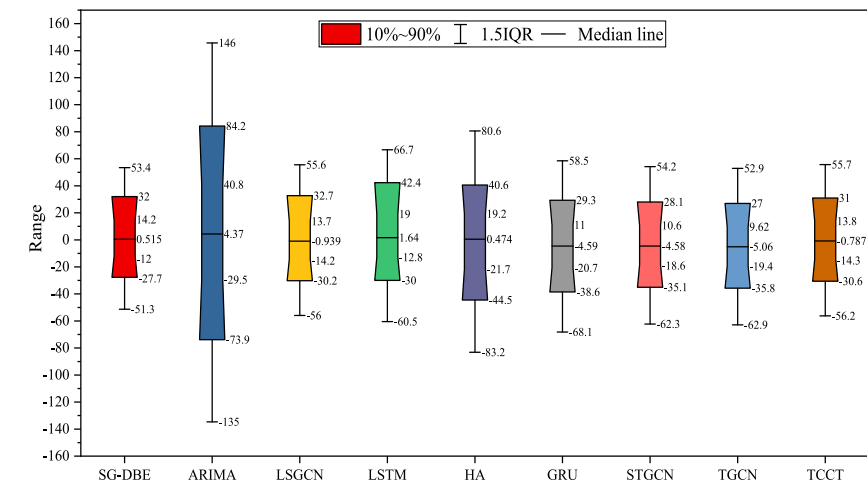


(a) Morning peak hours

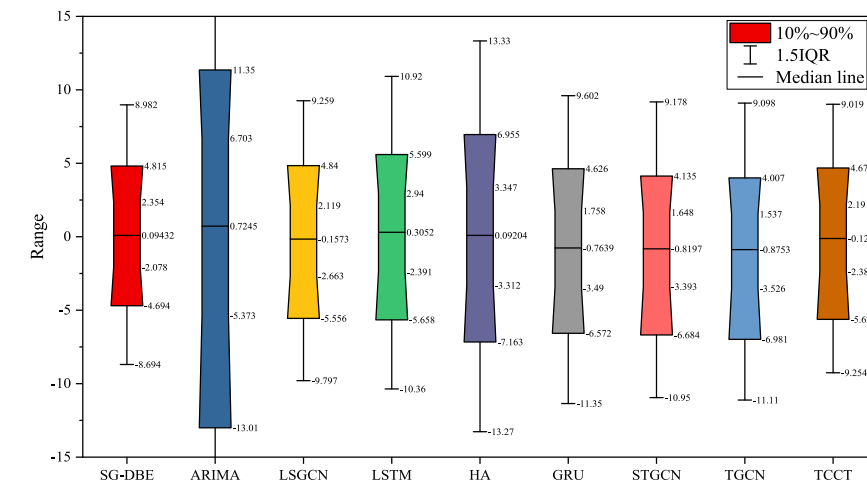


(b) Evening peak hours

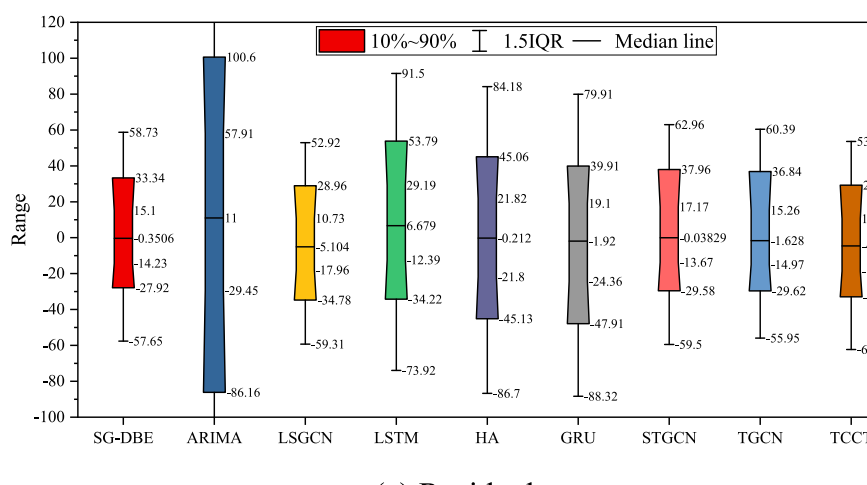
Fig. 13. Global Moran's I during peak hours.



(a) Residual

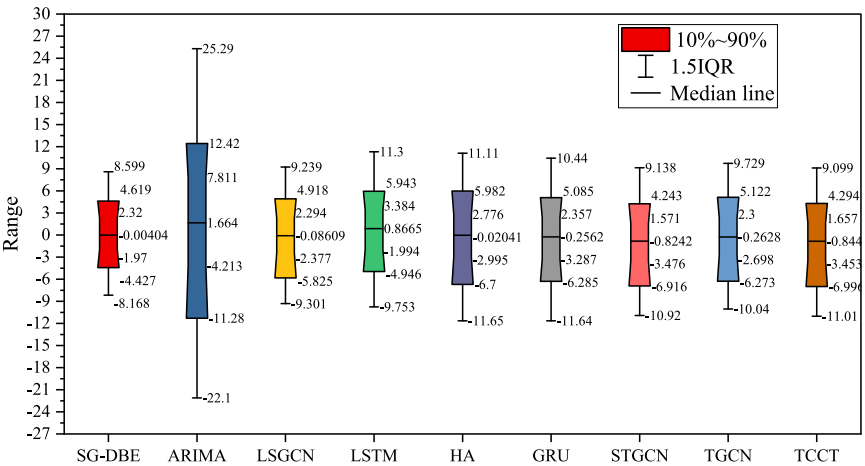


(b) Relative error

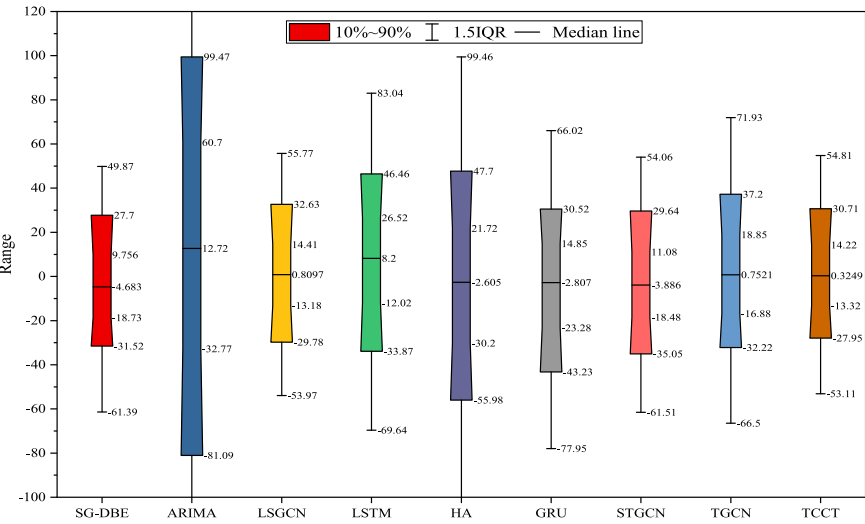


(c) Residual

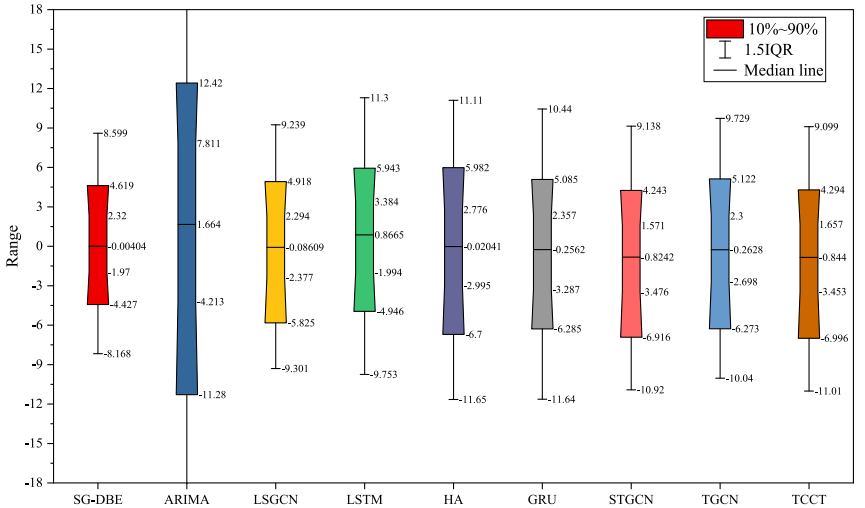
Fig. 14. Error statistics for different kinds of models.
((a)–(b): Whole day, (c)–(d): morning peak hours, (e)–(f): evening peak hours).



(d) Relative error



(e)Residual



(f) Relative error

Fig. 14. (continued).

Table 5
Details of SG-DBE with different combinations of input features.

	Method	Description
1	SG-DBE (A)	The overall network structure is the same as SG-DBE but with only input feature of crowd flow data (A)
2	SG-DBE (A + B)	The overall network structure is the same as SG-DBE but with input feature of crowd flow data (A) and dynamic influence of built environment (B)
3	SG-DBE (A + D)	The overall network structure is the same as SG-DBE but with input feature of crowd flow data (A) and static influence of built environment (D) (e.t. floorage)
4	SG-DBE (A + C)	The overall network structure is the same as SG-DBE but with input feature of crowd flow data (A) and data of time interval attribute (C)
5	SG-DBE (A + D + C)	The overall network structure is the same as SG-DBE but with static influence of built environment (D) (e.t. floorage) rather than the dynamic.

Table 6
Result of SG-DBE with different combinations of input features.

Model	MAE	MAPE	ATR(5;10)
SG-DBE (A)	19.81 (−8.73 %)	3.25 % (−9.43 %)	79.77 %; 96.02 %
SG-DBE (A + B)	18.69 (−2.58 %)	3.02 % (−1.68 %)	81.68 %; 96.82 %
SG-DBE (A + D)	19.02 (−4.39 %)	3.19 % (−7.40 %)	81.02 %; 96.42 %
SG-DBE (A + C)	19.42 (−6.59 %)	3.21 % (−8.08 %)	80.07 %; 96.30 %
SG-DBE (A + D + C)	18.72 (−2.74 %)	3.14 % (−5.72 %)	81.03 %; 96.48 %
SG-DBE	18.22	2.97 %	82.03 %; 97.19 %

Table 7
Result of three baseline models with dynamic influence of building.

Model	MAE	MAPE
LSTM(A + B)	21.18(+0.18 %)	3.55 %(+0.83 %)
T-GCN(A + B)	19.96(+2.40 %)	3.28 %(+4.57 %)
STGCN(A + B)	20.15(+3.96 %)	3.30 %(+2.36 %)

except our model. For 30 min time intervals, our SG-DBE still achieves a lower MAE than other deep learning-based models, only obtaining a little higher MAE than LSGCN). The experimental outcomes validate that our model excels in spatial-temporal feature extraction, thereby enabling more precise crowd flow predictions than traditional machine

learning and deep learning-based methods.

5. Discussion

5.1. Impact of input feature length parameters

We test the how length of input feature influences prediction by changing the steps of short-term and long-term data. The results (shown in Fig. 19) indicate that the MAE and MAPE show a decreasing trend and ATR shows an increasing trend as the step increases but respectively reaching a low peak and peak at a step of 10. Besides, When the long-term data step is set to 7, the overall MAE, MAPE and ATR are relatively considerable, compared to taking steps of 5 and 3. Those verifies that as the length of input feature influences, the prediction accuracy improves with the neural network can capture more information. However, when a neural network has to handle excessive of input feature, it increases its complexity and computational burden, potentially leading to the neural network not being able to fully learn from the training data or resulting in overfitting and finally prediction accuracy getting worse.

5.2. Impact of parameters

To verify the impact of different parameters on model performance, the numbers of RGACNN module, size of convolution kernel and dilation factor in TCN, size of multi-head mechanism in GAT, and layers of GRU separately were changed.

- Numbers of RGACNN module. The efficiency of deep learning is significantly influenced by the depth of the network. Thus, we deepened the network depth by setting multiple RGACNN modules at 1, 2, 3, 4 to explore the Impact of depth on prediction. The results are shown in Fig. 20(a) The performance of the model is the best with the numbers of RGACNN set at 1. Then, the values of MAPE and MAE increase from 2.87 to 4.03 and from 18.11 to 22.34 as the number of RGACNN blocks increases from 1 to 4. This is due to the stacking of the RGACNN modules, which makes the network much deeper, and the training process becomes more complex. Over-processing of the data leads to a decline in model performance.

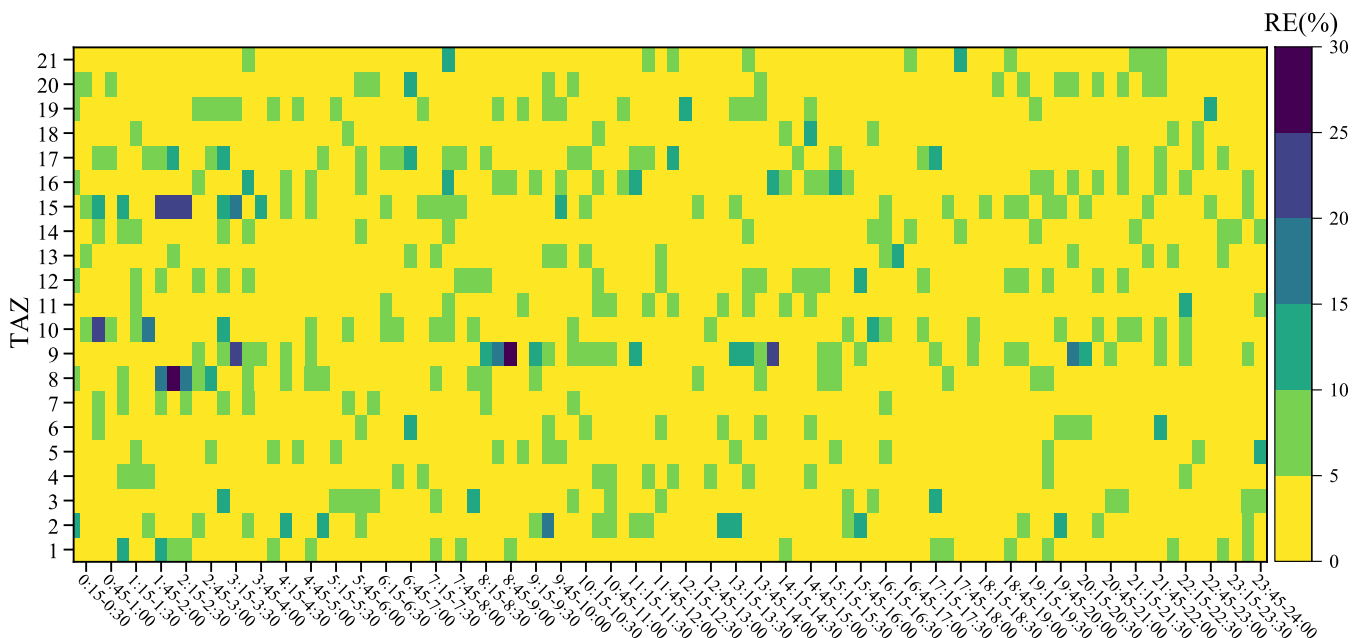
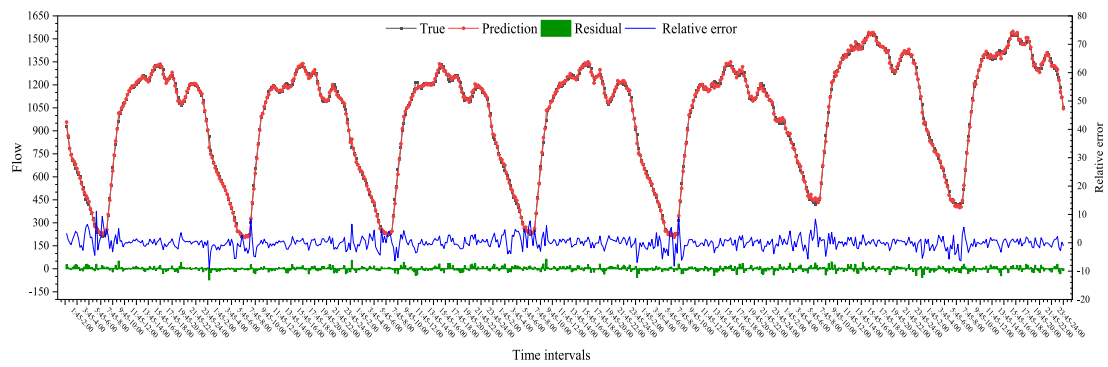
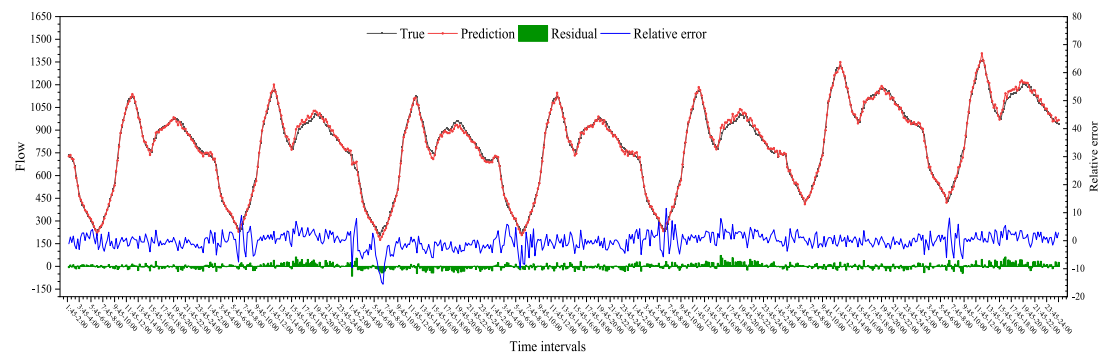


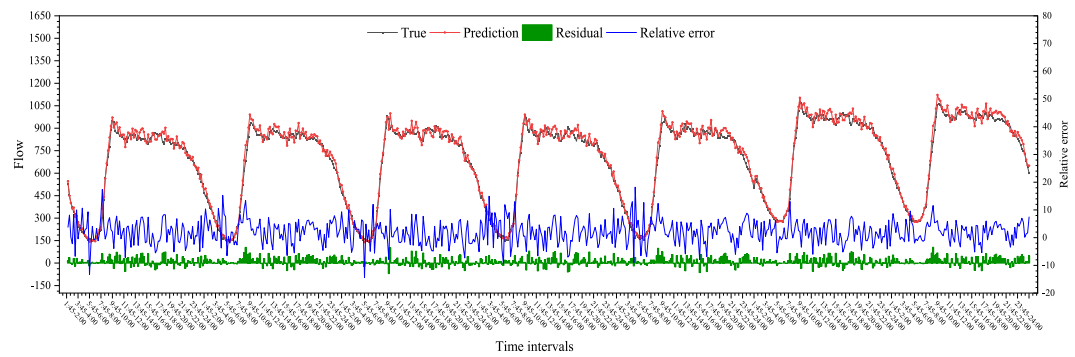
Fig. 15. Temporal distribution to relative errors of all TAZ.



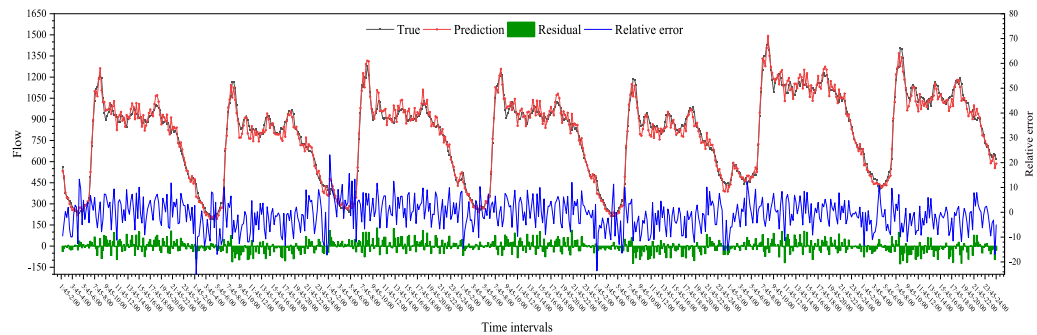
(a) 1st TAZ



(b) 3rd TAZ

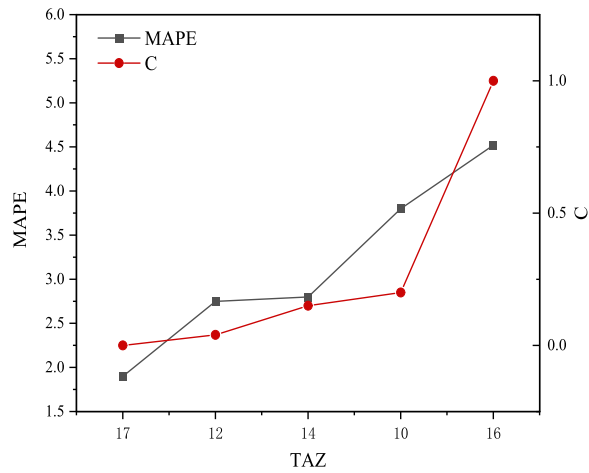


(c) 15th TAZ

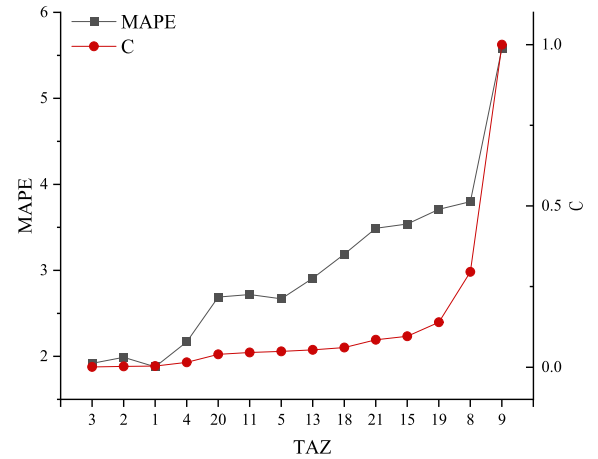


(d) 9th TAZ

Fig. 16. Error for crowd flow prediction in 4 TAZs.

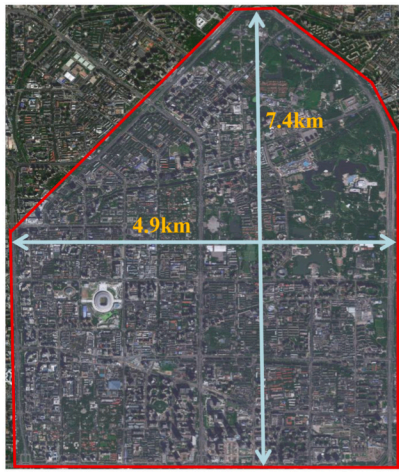


(a) Category A



(b) Category B

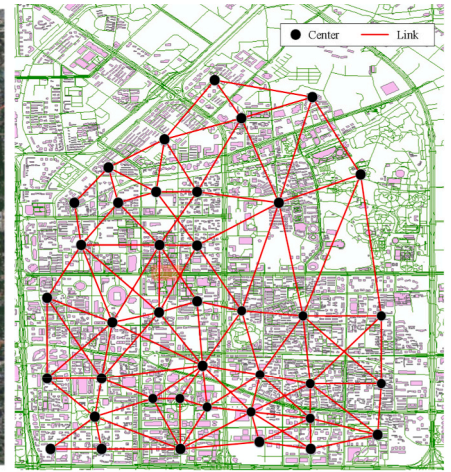
Fig. 17. Error analysis under the influence of time series volatility randomness.



(a) Size of study area



(b) Illustration of all TA



(c) Graph structure of study area

Fig. 18. Information of study area.

b) Size of convolution kernel and dilation factor in TCN. The temporal convolution kernel sizes are set from 2 to 6 to identify optimal kernel size. It can be seen in Fig. 20(b) that although larger convolution kernels can capture longer time series information, increasing the model's ability to perceive past crowd flow information, more parameters are introduced, which results in more complexity of the model. And the best prediction performance is when the convolution kernel size is 3. It suggests in Fig. 20(c) that the dilation factor set 2 is optimal, and shows that convolutional kernels can cover longer time spans, helping the model learn dependencies over longer distances in time series but leading to complexity.

Table 8
Overview of benchmark datasets.

Features		
1	Location	Beijing
2	Type of data	Taxi (GPS)
3	Time span	7/1/2013–10/30/2013
4	Time interval	15 min/30 min
5	Average sampling rate (s)	60

c) Size of multi-head mechanism in GAT. The size of multi head mechanisms determines the complexity of attention and the expressive power of the model. To determine optimal size of multi-head, the multi-head mechanism is set from 1 to 6. The result shown in Fig. 20(d) displays that increasing the size of multi-head mechanisms can strengthen the learning and expression abilities of the model thereby improving the performance of the mode.

Table 9
Evaluation of crowd-flow prediction on TaixBJ.

Model	MAE		MAPE	
	15 min	30 min	15 min	30 min
HA	54.40	68.11	10.12 %	13.62 %
ARIMA	48.71	57.18	8.58 %	11.33 %
T-GCN	11.25	12.64	4.43 %	5.34 %
LSTM	12.24	13.18	5.12 %	5.51 %
GRU	11.93	13.03	4.65 %	5.29 %
STGCN	11.21	12.68	4.58 %	5.69 %
LSGCN	10.72	12.18	4.26 %	4.57 %
TCCT	10.93	12.41	4.33 %	5.01 %
SG-DBE	10.44	12.20	4.05 %	4.55 %

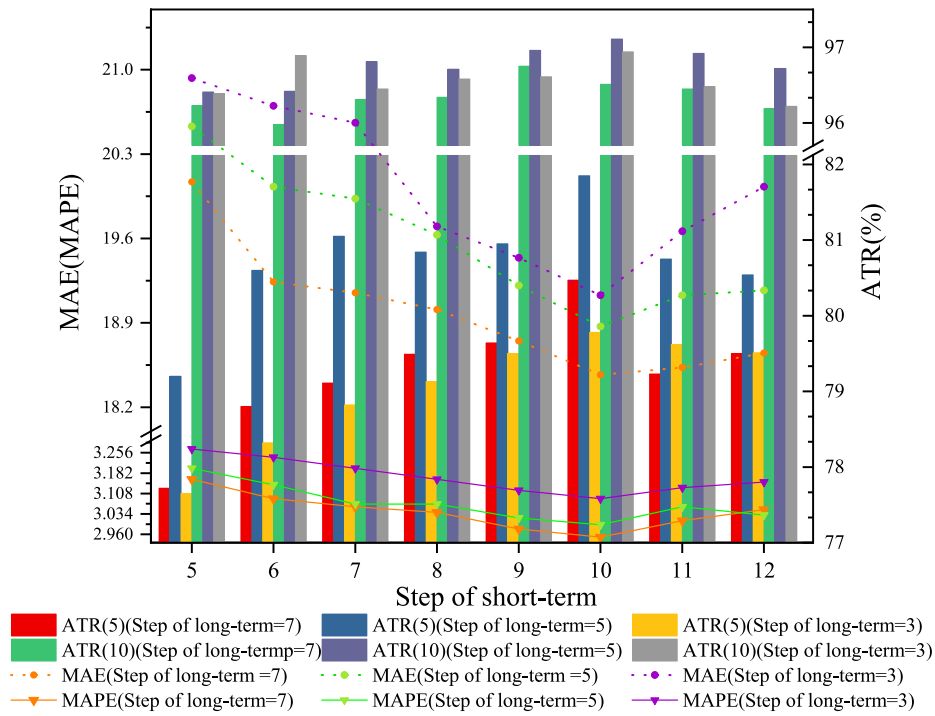


Fig. 19. Error for prediction in different feature length parameters.

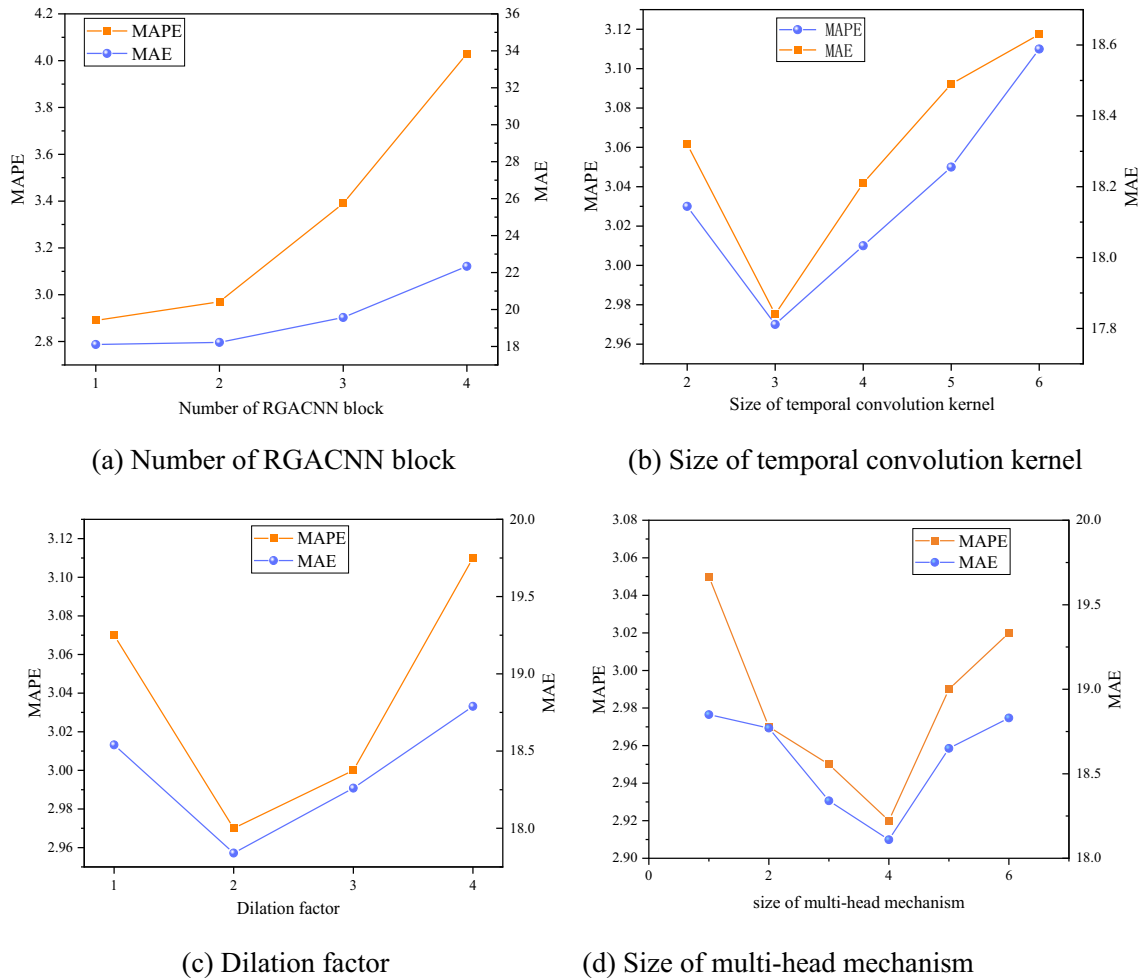


Fig. 20. Errors under various parameter settings.

Table 10

Details of RGACNN with different combinations of network layers.

	Method	Description
1	RGACNN (no TCN)	The overall network structure is the same as RGACNN but without TCN layer.
2	RGACNN (no GRU)	The overall network structure is the same as RGACNN but without GRU layer.
3	RGACNN (LSTM → GRU)	The overall network structure is the same as RGACNN but with GRU replaced by LSTM.
4	RGACNN (GCN → GAT)	The overall network structure is the same as RGACNN but with GAT replaced by GCN.
5	RGACNN (GRU ↔ TCN)	The overall network structure is the same as RGACNN but with swapping the order of GRU and TCN.

Table 11

Comparison of prediction performance between proposed model and its variants.

Model	MAE	MAPE	ATR(5;10)
RGACNN (no TCN)	19.33(−6.09 %)	3.22 % (−8.42 %)	79.30 %; 96.14 %
RGACNN (no GRU)	19.70(−8.12 %)	3.25 % (−9.43 %)	79.21 %; 96.10 %
RGACNN (LSTM → GRU)	18.46(−1.32 %)	3.05 % (−2.69 %)	81.72 %; 97.02 %
RGACNN (GCN → GAT)	19.16(−5.16 %)	3.12 % (−6.06 %)	80.87 %; 96.97 %
RGACNN (GRU ↔ TCN)	19.20(−5.38 %)	3.10 % (−4.37 %)	79.45 %; 97.11 %
RGACNN	18.22	2.97 %	82.03 %; 97.13 %

However, an overly large size of multi-head mechanisms results in overly complex input explanations and reducing generalization ability.

5.3. Impact of composition to RGACNN

To verify the advantages of the combined effect of two different mechanisms for capturing regularity in time sequences, several application changes based on RGACNN listed as follows are also constructed and tested. Details are shown in Table 10.

In Table 11, we can find that the performance of RGACNN decreased by 8.42 % and 9.43 % in MAPE, 6.09 % and 8.12 % in MAE on the condition of the RGACNN (no TCN) and RGACNN (no GRU). It proves the effectiveness of utilization of two mechanisms for extracting temporal dependencies within the time sequence. MAPE and MAE increase by 6.06 % and 5.16 % when GAT is replaced by GCN in RGACNN, testing the utility of considering the weight between the central node and adjacent nodes during capturing the spatial dependences. Interestingly, The RGACNN exhibits a relatively high MAPE, but the ATR (10) shows a comparatively higher value, while the ATR (5) is relatively lower, indicating that the RGACNN (GRU ↔ TCN) has a weaker capacity to minimize errors at the extremes but possesses a more stable error control. This is attributed to the TCN's extensive extraction of temporal information through dilated convolution, followed by the GRU's gating and recursive mechanisms to filter useful information, thereby maintaining stable model performance. The overall performance of the RGACNN is better than that of the RGACNN (GRU ↔ TCN), due to its initial use of GRU for information screening and utilization to the information remained by the TCN to enhance the model's performance.

6. Conclusions

Crowd flow possesses spatial-temporal patterns and thus shows potential predictability. Deep learning takes on performance than traditional models in prediction on crowd flow, benefited by strong

capability to identify spatial or temporal dependencies. In addition, this study demonstrates that the built environment can significantly improve accuracy of crowd flow prediction. Some studies characterize the impacts of the built environment as fixed attributes and employ neural networks to implicitly extract these characteristics. While this approach has enhanced predictive accuracy, it fails to account for the dynamic spatial-temporal influences of the built environment. To solve the issue, this study proposes a deep learning model based on sequence and graph considering dynamic influence of built environment (SG-DBE) for crowd flow prediction. The time-based multiscale geographically weighted regression model in SG-DBE is used to explicitly capture the dynamic influence of the built environment (BE) on crowd flow. A RGACNN module (coupling with GRU, GAT and TCN) is designed to capture the comprehensive spatial and temporal dependency of crowd flow and the BE dynamic influence. The validity of the model was demonstrated using the cases in Guangzhou. The proposed SG-DBE performs better with the MAPE of 2.97 % and MAE of 18.22. What's more, the comparison results show: 1) The prediction accuracy is improved through the dynamic impact of built environment considered. 2) The combined effect of two different mechanisms to capture temporal dependencies improves the model performance. To further verify the method's applicability to other datasets, we replicated the experiment on a public dataset (TaxiBJ dataset) in a specific area of Beijing. The results show that: 1) For 15 min time intervals prediction, our method reduces by 4.92 % MAE and by 2.62 % than the LSGCN model which performs the best among the eight models except our model; 2) For 30 min time intervals prediction, our method still achieves a lower MAE than other deep learning-based models, but a little higher MAE than LSGCN.

This method enables urban managers to enhance their perception of crowd flow conditions and make significant deployments. For instance, it allows for the implementation of control measures in areas with high pedestrian traffic, thereby preventing or mitigating the risks and impacts of traffic congestion, as well as potential risks to public safety. Additionally, those crowd flow conditions can be utilized to assist in identifying hotspots of urban traffic pollution, such as carbon emissions and noise.

Continuously improvement of the RGACNN module in SG-DBE is the future work. Such as considering the incorporation of attention mechanisms within temporal feature extraction modules, or the utilization of dynamic graph neural networks to better identify the spatial-temporal variability of nodes.

CRediT authorship contribution statement

Haibo Wang: Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Zhaolang Wu:** Writing – original draft, Validation, Software, Investigation. **Qing Hou:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Ming Cai:** Writing – review & editing, Project administration, Data curation.

Declaration of competing interest

The authors declare they do not have any known financial or personal relationships that could have been perceived to influence the findings presented in this paper.

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Appendix A. Parameter details of baseline model

Table 12
LSTM model parameter table.

Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of LSTM blocks	3
Hidden units	16, 48, 96
Training set: test set: validation set	6:1:3
Optimizer	Adam
Loss	MAE
Epoch	1000
Batch size	32
Learning rate	0.001

Table 13
GRU model parameter table.

Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of GRU blocks	3
Hidden units	16, 48, 96
Training set:test set:validation set	6:1:3
Optimizer	Adam
Loss	MAE
Epoch	1000
Batch size	32
Learning rate	0.001

Table 14
T-GCN model parameter table.

Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of T-GCN blocks	2
Kernel size of graph convolution	3
hidden units of T-GCN blocks	100
Training set:test set:validation set	6:1:3
Optimizer	Adam
Loss	MAE
Epoch	1000
Batch size	32
Learning rate	0.001 with a decay rate of 0.7 after every 100 epoch

Table 15
STGCN model parameter table.

Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of spatio-temporal convolution blocks	2
Kernel size of spatial convolution	3
Kernel size of temporal convolution	3
Channel configs of ST conv blocks	[[1, 32, 64], [64, 32, 128]]
Training set:test set:validation set	6:1:3
Optimizer	Adam
Loss	MSE
Epoch	200
Batch size	32
Learning rate	0.001 with a decay rate of 0.7 after every 100 epoch

Table 16
LSGCN model parameter table.

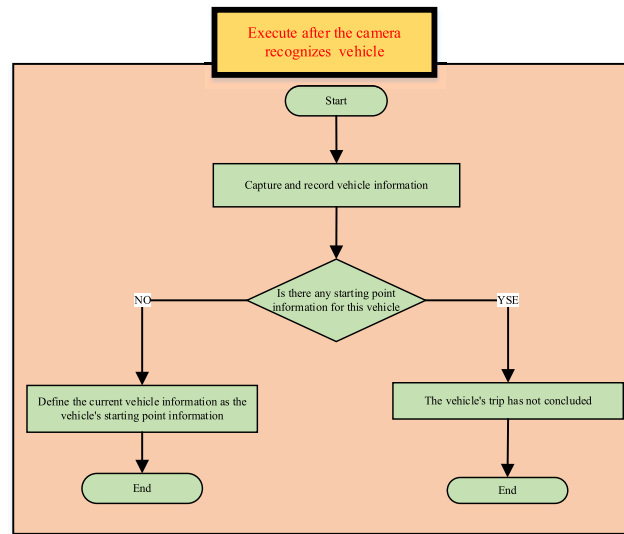
Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of LSGCN blocks	2
Graph convolution kernel size	3
GLU convolution kernel size	3
Channels of the first GLU, GCN, cosAtt, the second GLU	32, 32, 32 and 64
Training set:test set:validation set	6:1:3
Optimizer	Adam
Loss	MSE
Epoch	1000
Batch size	32
Learning rate	0.001 with a decay rate of 0.7 after every 100 epoch

Table 17
TCCT model parameter table.

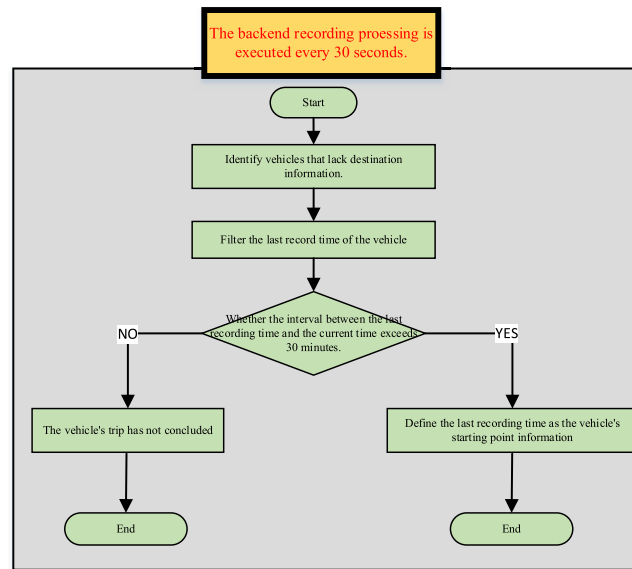
Name	Value
Number of nodes	21
Input timestep	16 (long-term: 7 + short-term: 9)
Output timestep	1
Number of LSGCN blocks	2
Dilated causal convolution	2
Convolution kernel size	3
Self-attention block	3
Transformer block	6
Training set:test set:validation set	6:1:3
Optimizer	Adam
Loss	MSE
Epoch	1000
Batch size	32
Learning rate	0.001 with a decay rate of 0.7 after every 100 epoch

Appendix B. Description on the data collection and preprocessing

The Fig. 21 illustrate the flowchart for identifying and defining vehicle trips. The process is divided into two modules, with the Fig. 20(a) representing the camera capture module and the Fig. 20(b) representing the backend processing module. Once a vehicle is detected by the camera, the camera capture module is immediately executed to record the vehicle's information, including the record time, intersection ID, camera ID, and vehicle number, etc. It then checks whether there is existing starting information (starting information from a trip already completed will not be reused for this judgment). If no starting information exists, this vehicle record is considered as the starting information; otherwise, the vehicle is considered to be in transit. The backend processing module is executed every 30 s, and if no further records of a vehicle are found after 30 min, it is assumed that the vehicle has completed its trip, and the last information of the vehicle is regarded as the destination information. Ultimately, we consider trips where both the origin and destination are out of the study area as pass-through trips. Such trips are neither attracted to nor repelled by the buildings, nor do they fall within the definition of inflow and outflow volumes. Therefore, these trips are not taken into account and are not included in the statistics.



(a) Flowchart for identifying



(b) Flowchart for defining vehicle trips

Fig. 21. Flowchart for identifying and defining vehicle trips.

Appendix C. Structure of three baseline model with dynamic influence of building

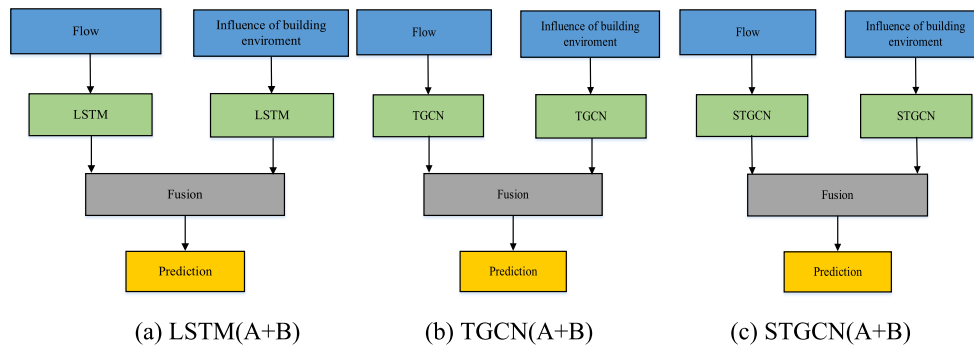


Fig. 22. Structure of three baseline model with dynamic influence of building.

Data availability

Data will be made available on request.

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